Shape bias is a readout property: a developmental two-stream model, frozen networks and human cortex

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Abstract

Humans name objects by shape when shape and texture disagree, yet the ventral visual stream is well described by texture statistics. We test whether this is a contradiction with a two-stream network approximating the magnocellular and parvocellular inputs: one ResNet-18 receives the luminance channel of a cone-opponent image under a developmental visual diet, the other its L−M channel. The streams end on opposite sides of the shape–texture boundary: the magno stream's shape bias is supplied by its input and survives training only under the diet; the parvo stream's texture bias is learnt. Fitted to eight Natural Scenes Dataset subjects, the texture-biased stream carries every extrastriate region. Behaviour, reading out from both, is shape-biased; one readout weight moves the rate from 0.40 to 0.90, and the weight that best predicts extrastriate cortex (0.33) is not the one that best predicts human choices (0.67). In ten frozen networks without a stream split, including ImageNet models, a gain on their own shape-leaning units moves the rate monotonically and reorders the field's ranking at no accuracy cost, and in our single trunks extrastriate cortex again loads on the texture-leaning units. Shape-biased behaviour does not require shape-biased representations; the cue-conflict rate is a readout statistic.

1. Introduction

Humans presented with an image whose shape and texture point to different categories name the shape: on the cue-conflict images of Geirhos et al. (2018a), ten observers did so on 96% of decided trials, while ImageNet-trained convolutional networks named the texture. Closing that gap has become a standard target for biologically motivated models (Hermann et al., 2020; Jang & Tong, 2021), and a developmental visual diet — training on input whose acuity, contrast sensitivity and chromatic sensitivity follow the human developmental trajectory — is among the most effective routes to it (Lu et al., 2026).

A second body of evidence sits uneasily beside the first. Along the ventral stream, object representations are well described by texture statistics: category-selective human cortex fails to discriminate natural images from feature-matched scrambles that preserve texture statistics while destroying spatial arrangement, yet still discriminates between categories (Jagadeesh & Gardner, 2022); macaque V4 responses to natural textures are predicted by their image statistics (Okazawa et al., 2015); and in inferotemporal cortex texture is the more decodable cue (Jagadeesh & Livingstone, 2024; Komatsu & Goda, 2018). The premise is scoped: the human evidence is one study with one manipulation, the rest is macaque, and "texture-like" means that a texture-statistics description suffices, not that shape is absent. The argument needs only that the ventral description is substantially texture-like.

If behaviour is shape-biased and the ventral stream texture-biased, then either the two measures disagree about what shape bias is, or behaviour is not a readout of one representation. The second horn is a hypothesis about architecture — other pathways contribute to the behavioural bias — that models of ventral-stream computation, which almost universally take a single RGB image into a single trunk, cannot express. Two facts about the primate visual system make it concrete.

The visual system does not receive one image: it receives a luminance-dominated, spatially coarse magnocellular signal and a chromatically opponent, finer parvocellular one, separated before cortex and later recombined (Livingstone & Hubel, 1987; Kaplan & Shapley, 1982; Derrington et al., 1984; Nassi & Callaway, 2009; Ferrera et al., 1994). And that input is not stationary over development: acuity and contrast sensitivity improve over years, so the system first learns from a low-passed world. The two-stream hypothesis (Mishkin et al., 1983; Goodale & Milner, 1992) and the emergence of ventral- and dorsal-like streams in a two-pathway network trained on one objective (Bakhtiari et al., 2021) motivated a two-stream architecture; here the streams are defined by their subcortical-like inputs rather than by their cortical targets.

This paper asks two questions, in the order the originating study posed them (Manocchio, 2026). Can two streams fed magnocellular- and parvocellular-like inputs diverge in their internal representations while optimising a single objective and producing shape-biased behaviour? And does such a system say anything about why the ventral stream can be texture-like while behaviour is shape-biased? We build the system — two ResNet-18 trunks receiving the luminance and L−M channels of a cone-opponent image, with a developmental diet on the luminance channel (Fig. 1) — and examine it with linear probes and cue-conflict behaviour on the streams (Fig. 2), voxel-wise encoding models of eight subjects' 7T fMRI responses to natural scenes (Fig. 3), and the behavioural readout that combines the streams (Fig. 4). The streams diverge, by different routes; the texture-biased one carries extrastriate cortex; and the integrated behaviour is shape-biased because a behavioural rate is a property of a readout over components, not a report of any one of them — a property that holds, in kind, in networks with no stream split at all (Fig. 6). Because the account predicts that properties bundled with shape bias need not travel with it, we test that too, and find that robustness, agreement with human decisions and representational dimensionality all come apart from the rate (Fig. 5).

Relative to the originating thesis, three things changed. Every condition is now trained with three seeds and every claim states its replication unit; the representational-similarity analysis that had found no stream-to-brain mapping is replaced by variance partitioning of encoding models, which removes the global stream preference that had defeated it; and the stream weighting that the thesis fixed by hand is swept, so that the dependence of behaviour on the readout is measured rather than assumed.

2. Results

2.1 The model, and what the diet costs

The network splits its input once, before any convolution (Fig. 1a). Each image is converted to the DKL cone-opponent space; stream 1 receives the luminance channel, stream 2 the L−M channel, and each is a ResNet-18 trunk. Under the developmental visual diet the luminance channel passes through a Gaussian blur ($\sigma = 4$ px) and the scale-free contrast-sensitivity cut of Lu et al. (2026) at a fixed age of 48 months, so that the transform is separated from the trajectory (Methods 4.1). The trunks are initialised bit-identically, and a control in which both streams receive the same image leaves all 120 tensor pairs identical after 150 epochs, so every stream difference below is attributable to the inputs. A linear classifier reads out $w \cdot y_1 + (1-w) \cdot y_2$ of the two streams' penultimate features, with $w = 0.5$ during training.

Four front-end conditions are used throughout (Fig. 1a, table): the diet; the no-diet control; a channel-swap control in which the identical transform is applied to the L−M channel instead; and a `parvo_highsf` control in which stream 2 additionally receives untransformed luminance, because real parvocellular cells carry both. Networks were trained on MiniEcoset (112 classes;

Thorat & Kietzmann, 2024) for 150 epochs, three seeds per condition (Methods 4.2).

The diet costs accuracy (Fig. 1c). The no-diet conditions converge to 0.70 validation top-1; the diet arm reaches 0.556 at its best saved checkpoint against 0.701 and, uniquely, degrades after its peak, ending at 0.445 (range 0.347–0.527). Because that degradation would otherwise handicap the diet arm, behavioural and out-of-distribution results use each seed's best saved checkpoint, stated wherever the policy matters (Methods 4.7). Two single-stream controls trained with the identical diet — a ResNet-18 (11.2M parameters) and a ResNet-34 parameter-matched to the two-stream model (21.3M against 22.4M) — reach 0.599 and 0.623, so the accuracy cost of the split is not a capacity artefact (Fig. 4c).

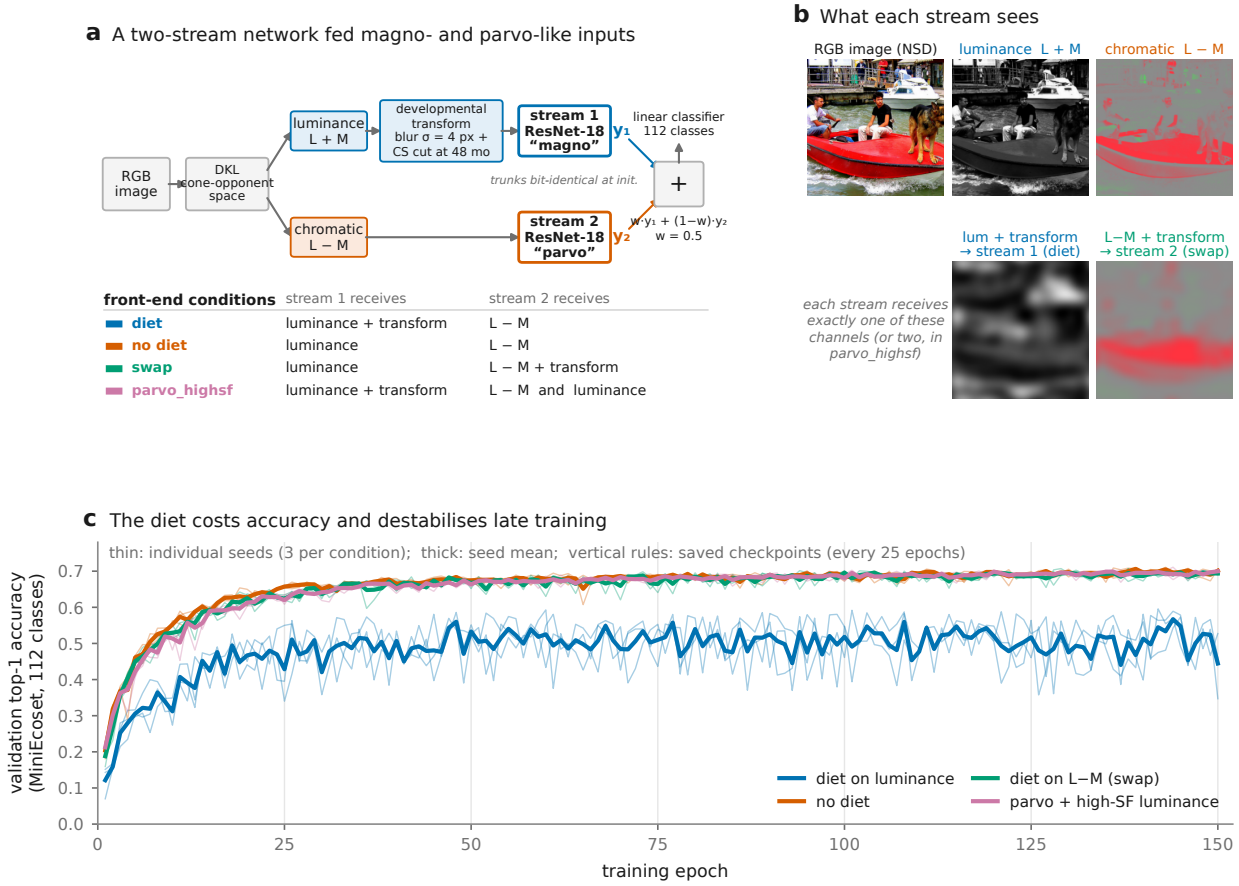


Figure 1 | A two-stream network fed magnocellular- and parvocellular-like inputs under a developmental visual diet.

(**a**) The model. Each RGB image is converted to the DKL cone-opponent space and split once, before any convolution: stream 1 ("magno") receives the luminance channel, stream 2 ("parvo") the L−M chromatic channel. Under the developmental visual diet the luminance channel additionally passes through a Gaussian blur ($\sigma = 4$ px) and the scale-free contrast-sensitivity cut of Lu et al. (2026), evaluated at a fixed age of 48 months. The two ResNet-18 trunks are initialised bit-identically, so any difference between the trained streams is attributable to their inputs. A linear classifier reads out the weighted sum $w \cdot y_1 + (1-w) \cdot y_2$ of the two streams' penultimate features, with w fixed at 0.5 during training. The table lists the four front-end conditions used throughout: the diet, the no-diet control, the channel-swap control (the identical transform moved onto L−M) and the `parvo_highsf` control (stream 2 additionally receives untransformed luminance).

(**b**) What each stream sees, for one NSD image (id 15504, the most chromatic of a fixed candidate grid), rendered with the exact training front end. Top: the RGB image and its luminance and L−M channels; bottom: the two channels after the developmental transform. The L−M channel is contrast-stretched for display only.

(**c**) Validation top-1 accuracy over 150 epochs of training on MiniEcoset (112 classes), three seeds per condition. Thin lines are individual seeds, thick lines the seed mean; vertical rules mark the epochs at which weights were saved. The diet costs accuracy — 0.556 against 0.701 at the best saved checkpoint — and, uniquely among the four conditions, degrades after its per-epoch peak (final-epoch mean 0.445, range 0.347–0.527). Every behavioural and cortical result in this paper states which checkpoint policy it uses.

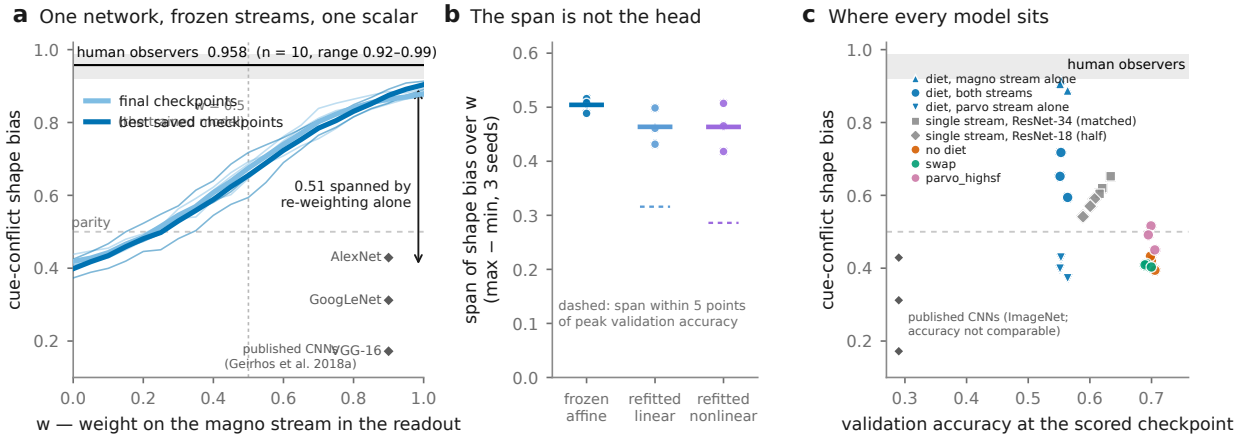


Figure 4 | Behaviour reads out from both streams, and the shape-bias rate is a property of that readout.

(**a**) Cue-conflict shape bias as a function of w, the weight on the magno stream in the readout full = classifier($w \cdot y_1 + (1-w) \cdot y_2$), at 21 values of w, for three diet-trained seeds under both checkpoint policies. Thin lines are seeds, thick lines the seed mean. Nothing is retrained, refitted or reselected along the x-axis: y_1 and y_2 do not depend on w, so the sweep moves the readout while both representations stay bit-identical. Shape bias runs from 0.40 to 0.90 at the best saved checkpoints; w = 0.5 is the trained model and reproduces its published per-seed values exactly. The ten human observers who viewed the identical images (mean 0.958, range 0.919–0.987; grey band) and the three CNNs of the Geirhos et al. (2018a) release are drawn on the same axis. No value of w reaches the lowest individual observer.

(**b**) The span is not a property of the affine head. Span of shape bias over w (max − min) per seed, best saved checkpoints, for the frozen affine head of (a), a linear head refitted on the frozen features at each w, and a two-layer nonlinear head refitted likewise (identical budget, same initialisation seed at every w). Median spans 0.508, 0.461 and 0.466; the refitted-linear and nonlinear heads agree to within 1%. Dashed bars: the span restricted to readouts within 5 points of peak validation accuracy (0.283 for the nonlinear head), the range over which the rate can be moved without buying it with accuracy.

(**c**) Where every model sits: cue-conflict shape bias against validation accuracy at the scored checkpoint, one marker per training seed. The diet-trained network's two streams read out alone bracket its integrated readout (0.903, 0.400 and 0.655); the two single-stream controls trained with the identical diet (one ResNet-18 trunk, 11.2M parameters; one ResNet-34 trunk, 21.3M, parameter-matched to the 22.4M two-stream model) are more accurate than the two-stream model and lie at 0.567 and 0.626. The three no-diet conditions cluster near parity at 0.70 accuracy. The published CNNs were trained on ImageNet and their accuracy is not on this axis; they are drawn at the left margin for their shape bias only.

2.2 The two streams specialise in opposite directions, by different routes

Read out separately with a linear probe on the cue-conflict images, the two streams of the diet-trained network end on opposite sides of the shape–texture boundary: shape − texture accuracy +0.819 for the magno stream and −0.518 for the parvo stream, means of three seeds (Fig. 2a). Behaviour agrees: read out alone, the magno stream's cue-conflict shape bias is 0.878 and the parvo stream's 0.418, against 0.675 for the integrated readout (Table 1). This is the divergence the thesis reported on a single model per condition, now replicated across seeds.

Table 1. The two streams end on opposite sides of the boundary, by different routes.

	magno stream (luminance + transform)	parvo stream (L−M)	integrated
linear probe, shape − texture (0 = sign flip), 3 seeds	**+0.819**	**−0.518**	—
cue-conflict shape bias (0.5 = parity), 3 seeds	**0.878**	**0.418**	**0.675**
linear probe, untrained, seed 0	**+0.796**	−0.087	—
linear probe, trained, seed 0	**+0.810**	**−0.482**	—
contribution of learning, seed 0	**+0.014**	**−0.395**	—

The decomposition arises asymmetrically. Scoring the same probe on the untrained network with the same front end separates two routes (Fig. 2a, Table 1). The magno stream's shape bias is essentially complete before any weight is updated — the low-pass and contrast-sensitivity transform does it — and training adds +0.014; the parvo stream begins barely biased and acquires its texture preference over training. The shape-biased component is supplied by the input; the texture-biased component is built by learning. What ordinary training does to the input's contribution is the decisive contrast: with the transform removed, the untrained luminance stream starts lower (+0.417) and training drives it across the sign flip to −0.226. The unblurred luminance stream does not merely fail to become shape-biased; it becomes texture-biased. The parvo stream ends texture-biased whether or not the other stream is dieted (−0.518 against −0.504).

Which property of the input supplies the shape bias? Decomposing the front end into two factors — which carrier a stream receives (achromatic luminance or chromatic L−M) and whether the developmental transform is applied to it — gives four cells, two of which the swap and `parvo_highsf` conditions provide (Fig. 2b, c). On the probe, the chromatic carrier gives −0.504 without the transform and −0.145 with it; the achromatic carrier gives −0.271 without and +0.819 with, seed ranges 0.014–0.043. Both single factors move a stream toward neutral; only the conjunction crosses zero, and it crosses all the way, by +0.936 with no seed overlap; cue-conflict shape bias on the same four cells replicates the interaction (Extended Data Table 4). The shape bias requires the developmental transform applied specifically to an achromatic channel — a relational fact about what a stream receives, not a property of a stream called "magno".

Two qualifications belong here. First, cue-conflict shape bias is computed only over images a readout decides as the shape or the texture category, a count that varies four-fold across cells (165 to 583 of 1,200; Fig. 2d), so the three non-conjunction behavioural cells are read as "no strong preference". The probe classifies every image and gives the same pattern, and so does a forced two-way estimator that scores every image by whether the shape-category logit exceeds the texture-category logit: it compresses values toward 0.5 but preserves every ordering in this paper (the four cells read 0.44, 0.51, 0.35 and 0.76; Extended Data Table 10). Second, the diet-trained models are less accurate. We pre-registered the fatal outcome — a no-diet model at the diet's accuracy with a shape-biased magno stream — and retrained the no-diet network with per-epoch checkpointing, selecting by rule the epoch nearest the diet's accuracy (0.548 against 0.556): matching accuracy closes 7.6% of the probe gap and 7.0% of the behavioural gap, and the matched magno stream stays on the texture side (Fig. 2e, f; n = 1).

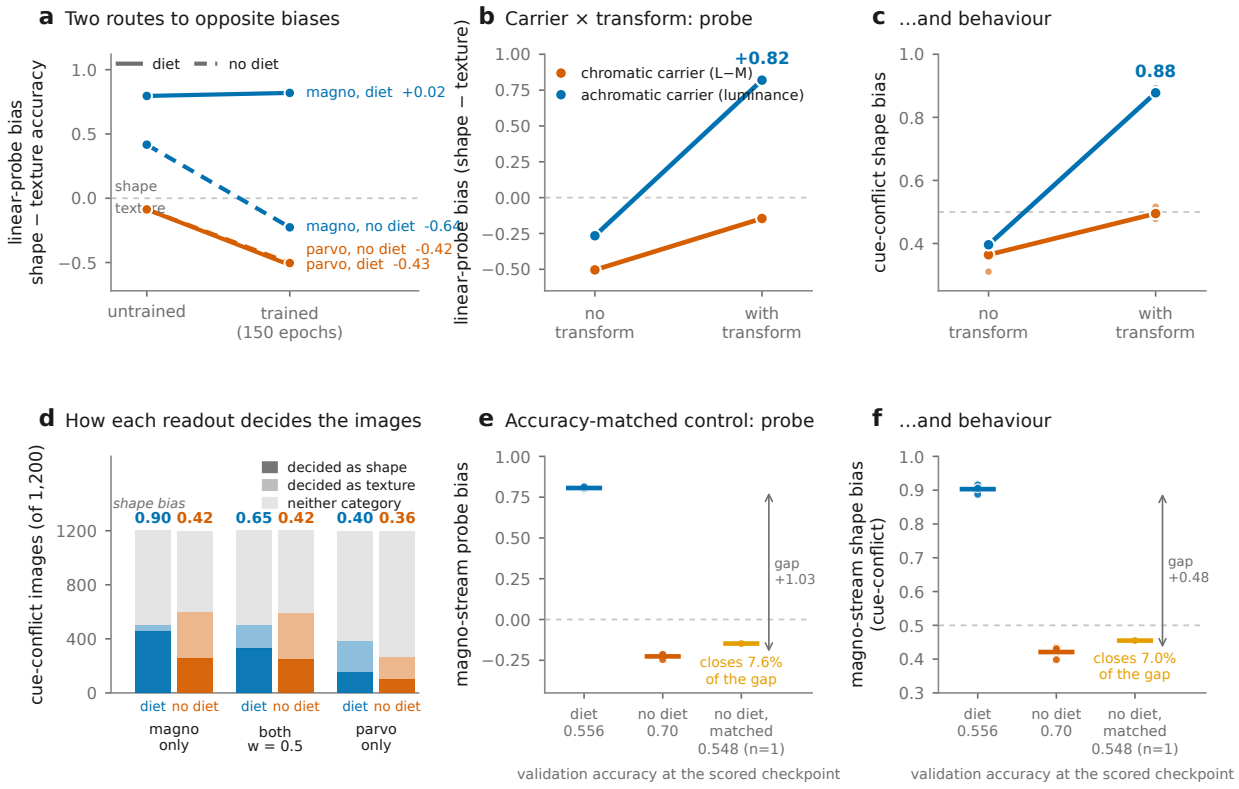


Figure 2 | The two streams end training on opposite sides of the shape–texture boundary, and they get there by different routes.

(**a**) Linear-probe shape−texture bias of each stream's features (shape-label accuracy minus texture-label accuracy on the 1,200 Geirhos cue-conflict images; zero is the sign flip), before and after training, with the diet (solid) and without it (dashed). The untrained references are single models with no training seed; trained points are the mean of three seeds with the seed range as a bar. Under the diet the magno stream is already shape-biased before any weight is updated (+0.80) and training adds +0.02; the parvo stream starts near neutral and acquires its texture bias over training (−0.43). Without the diet, training drives the luminance stream from +0.42 across the sign flip to −0.23: ordinary training destroys the shape bias the input supplies. The parvo stream ends texture-biased whether or not the other stream is dieted.

(**b, c**) The carrier × transform 2×2 on the probe (b) and on cue-conflict shape bias (c). Each line is one carrier (chromatic L−M or achromatic luminance); the x-axis is whether the developmental transform is applied to it. Points are training seeds (three, or two for the achromatic/no-transform cell); the thick line is their mean. Neither factor alone crosses the boundary; only the conjunction does, on both instruments. Cells are read out on stream 2 in three cases and on the diet's magno stream in the fourth (Methods).

(**d**) How each readout decides the 1,200 cue-conflict images at the best saved checkpoint (diet) and the final checkpoint (no diet): images decided as the shape category, as the texture category, or as neither of the two. Shape bias is shape/(shape + texture); the number of decided images varies about four-fold across readouts, so rates near parity are read as "no strong preference". A forced two-way estimator that keeps every image (shape logit against texture logit) compresses all rates toward 0.5 but preserves every ordering shown here (Extended Data Table 10).

(**e, f**) Accuracy-matched control. The diet-trained models are less accurate, so the conjunction effect could in principle be an accuracy effect. A no-diet network retrained with per-epoch checkpointing and stopped at the epoch nearest the diet's accuracy (epoch 10, 0.548 against 0.556) closes 7.6% of the probe gap (e) and 7.0% of the behavioural gap (f), and its magno stream stays on the texture side. This control rests on one matched run (n = 1); the diet and no-diet arms show three seeds each.

2.3 In cortex, the texture-biased stream carries the ventral end of the hierarchy

The thesis had asked whether the two streams align with different cortical streams, using representational similarity analysis (Kriegeskorte et al., 2008) against seven regions of the Natural Scenes Dataset, and found that whichever stream was the more predictive correlated better with every region. We replicated that result, understood it as a general-factor problem — a correlation cannot separate a global stream preference from regional structure — and replaced the instrument: voxel-wise banded-ridge encoding models for each of the eight NSD subjects, one feature band per stream, with every voxel's cross-validated explained variance partitioned into the part only the magno band accounts for (U_m) and the part only the parvo band does (U_p) (Methods 4.5). The stream-preference profile — the per-voxel index $(\mathrm{U_m} - \mathrm{U_p})/(\mathrm{U_m} + \mathrm{U_p})$, median over reliable voxels per ROI — across V1v $\rightarrow$ hV4 is Fig. 3a.

The two trained models sit on opposite sides of the global stream-preference axis: without the diet every retinotopic area is magno-explained, with it every area is parvo-explained. That global difference is the general factor that defeated the correlational analysis, and it is not the result. The result is the shape of the profile. Only the diet arm descends steeply from V1 to hV4 — early visual cortex relatively more magno-explained, hV4 relatively more parvo-explained, the direction a luminance/low-spatial-frequency versus chromatic split predicts — and the descent belongs to the model whose global preference points the other way, which a model simply preferring one stream everywhere could not produce.

Quantified as the contrast hV4 − mean(V1v, V1d) of the per-ROI median (Fig. 3b; Extended Data Table 1), the gradient is -0.745 to -0.834 across the three diet-trained seeds against -0.049 to -0.096 without the diet, with the same separation on the raw statistic U_m − U_p; all six comparisons reach the exact Wilcoxon floor for eight subjects ($p = .0078$) with complete seed separation. The per-subject Spearman ρ between hierarchical rank and stream preference is -0.964 to -0.973 with the diet and -0.503 to -0.606 without (Fig. 3c; Extended Data Table 2). Because the rank vector [1, 1, 2, 2, 3, 3, 4] admits exactly 630 labellings of the seven ROIs, the permutation null is enumerable; it calibrates the extremity of a result that was not pre-specified — the gradient is the survivor of seven regional hypotheses, six of which failed (below) — and does not test a prediction. The diet arm's ρ is the single most extreme of all 630 in every seed and on both statistics ($p = 1/630 = .0016$); the no-diet arm does not reach that floor ($p = .019$ to $.092$), so we claim nothing there.

The diet does not create the gradient; it decides what training does to the one the input supplies. An untrained network on the diet front end already sits at -0.544 (8/8 subjects, $p = .0078$), and on the no-diet front end at almost the same value (-0.479), so the two front ends alone impose no different gradient. Training with the diet steepens it; training without it erodes it to near zero (Fig. 3b). Correcting each arm for the signed bias of the estimator itself — a simulated floor of the wrong sign, $+0.05$ and $+0.04$ — leaves a paired displacement of -0.871 with the diet against -0.130 without, each non-zero in every subject. The gradient survives the two front-end controls and replicates across codebases: an independently trained checkpoint from the thesis's own pipeline, with its progressive developmental schedule, gives -0.793.

The fitted models allow one more test that needs neither probe nor readout: pushed through models fitted on natural scenes only, the cue-conflict images yield a per-voxel predicted shape-bias index whose magno − parvo difference is $+1.53$ to $+1.57$ with the diet against $+0.08$ to $+0.11$ without (8/8 subjects, $p = .0078$ in every seed; Fig. 3d; Extended Data Table 3).

The gradient survives its nuisance axes and the instrument's own bias. Receptive-field size, eccentricity coverage and the noise ceiling all change from V1 to hV4, and the diet's magno stream is a low-pass instrument by construction. The diet gradient is negative within every one of NSD's

five eccentricity bands and every pRF-size tertile while the no-diet arm stays near zero in each; it grows with eccentricity (-0.26 in the fovea to -0.93 beyond 4°) but is present at matched eccentricity, the same dissociation holds on continuous pRF eccentricity and size within subject, and the estimator's own simulated bias is small, flat across ROIs and of the wrong sign (Extended Data Fig. 3; Extended Data Tables 8 and 9; the full account is Supplementary Note S3).

The gradient does not stop at hV4, and it does not turn. The stored fits cover every voxel of NSD's general visual mask, so the same index can be read on the regions the retinotopic hierarchy leads into (Extended Data Fig. 4; Extended Data Table 11). Under the diet the index runs from -0.05 in V1v through -0.81 in hV4 and then sits between -0.80 and -0.94 in every extrastriate region — ventral VO1/2 and PHC1/2, lateral LO1/2 and TO1/2, dorsal V3A/B and IPS0–2, NSD's stream parcels and the fLoc face, place, body and word regions alike. Every region-minus-V1 contrast is negative in 8/8 subjects (ventral -0.76, lateral -0.75, dorsal -0.75; $p = .0078$ each) and no region-minus-region contrast departs from zero under either arm (dorsal − ventral $+0.015$ with the diet, $+0.005$ without; all $p > .05$). The components show what saturates: with the diet the parvo band's unique variance rises from 0.10 in V1v to 0.26 in PHC2 while the magno band's collapses to about 0.01 beyond V3; without the diet the luminance stream rises over the same range and the L−M stream stays near 0.01 (Extended Data Fig. 4b). The texture-biased stream therefore carries extrastriate cortex as a whole, not the ventral end specifically, and the thesis's magno→dorsal, parvo→ventral mapping fails because the diet's magno stream contributes no unique variance anywhere beyond V3.

Three things this section does not settle are stated here rather than left to be found. The diet predicts BOLD worse, not better: median r^2 normalised by the noise ceiling is 0.405 against 0.464 without the diet, in 8/8 subjects (Fig. 3e), so the diet's case rests on stream differentiation and the gradient, not on prediction accuracy. The two front-end controls are degenerate on this instrument (swap: U_p = 0.006; `parvo_highsf`: U_m = 0.007), so the correct statement is that only the diet on luminance yields two streams that both carry substantial unique cortical variance and a gradient between them (Fig. 3a). And the gradient was not pre-specified: it is the survivor of seven regional hypotheses, six of which failed — including the thesis's magno→dorsal mapping, an hMT+ specificity and a foveal/peripheral contrast whose sign flipped when the blur was removed — reported because it alone survives removal of that manipulation, and read as sign and ordering rather than as a calibrated effect (Supplementary Notes S1 and S3; Extended Data Table 1 for the subject-01 influence check).

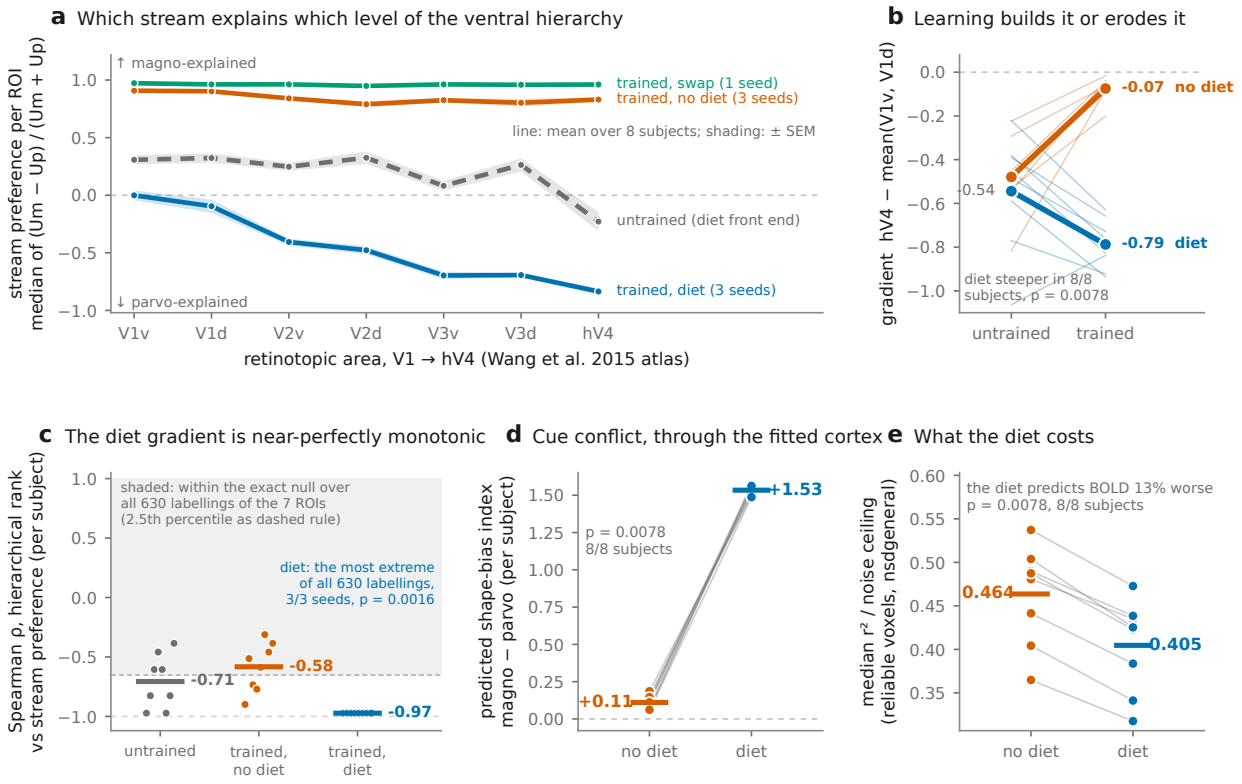


Figure 3 | In human visual cortex the texture-biased stream carries the ventral end of the hierarchy, and the diet decides whether learning builds that gradient or erodes it.

Voxel-wise banded-ridge encoding models were fitted on the Natural Scenes Dataset (8 subjects, `nsdgeneral` mask) with one feature band per stream, and every voxel's cross-validated explained variance was partitioned into the part only the magno band accounts for (U_m) and the part only the parvo band does (U_p). Stream preference per ROI is the median over reliable voxels of the per-voxel index (U_m − U_p)/(U_m + U_p) (noise ceiling > 0.1, U_m + U_p > 10^{-3}, at least 20 voxels); the raw statistic uses the per-voxel difference U_m − U_p.

(**a**) Stream-preference profile across the retinotopic hierarchy V1v → hV4 (Wang et al. 2015 atlas). Lines are means over the 8 subjects, shading ± SEM; the trained diet and no-diet arms average three training seeds per subject, the untrained network and the channel-swap control are single models. The two trained arms sit on opposite sides of the global stream-preference axis (no-diet magno-explained everywhere, diet parvo-explained), and only the diet arm descends steeply from V1 to hV4. The swap control leaves the parvo stream contributing almost no unique variance anywhere (median U_p = 0.006), so its flat profile reflects a degenerate decomposition rather than an absent gradient.

(**b**) The gradient hV4 − mean(V1v, V1d) per subject, from each arm's own untrained front end to the trained network (seed-mean per subject). The two untrained front ends start close together (−0.54 with the diet front end, −0.48 without); training with the diet steepens the gradient to −0.79, training without it erodes it to −0.07 (three seeds each: −0.834/−0.785/−0.745 against −0.079/−0.096/−0.049). The diet arm is steeper in 8/8 subjects (exact Wilcoxon p = .0078), and the same ordering holds on the raw statistic median(U_m − U_p).

(**c**) Monotonicity: Spearman ρ between hierarchical rank (V1 < V2 < V3 < hV4) and stream preference, per subject. The shaded region is the interior of the exact permutation null over all 630 distinct labellings of the seven ROIs; the dashed rule is its 2.5th percentile. The diet arm's observed ρ is the single most extreme of all 630 labellings in every seed (p = 1/630 = .0016); the no-diet arm does not reach that floor (p = .019 to .092).

(**d**) In-silico cue conflict. The 1,200 cue-conflict images pushed through the fitted encoding models yield a per-voxel predicted shape-bias index; plotted is the magno − parvo difference, median over reliable voxels, paired within subject. The effect is carried by the magno stream (+0.79 with the diet against −0.64 without);

the parvo streams are indistinguishable (-0.74 against -0.75).

(**e**) What the diet costs in prediction: median r^2 normalised by the noise ceiling over reliable voxels. The no-diet model predicts BOLD better in 8/8 subjects (0.464 against 0.405, a 13% cost). Panels (d) and (e) use seed 0 of each arm.

2.4 Behaviour reads out from both streams, and the rate is a property of the readout

The two streams are oppositely biased and the texture-biased one carries extrastriate cortex. What remains is what the behavioural readout does with them. The integrated readout's cue-conflict shape bias is 0.675 ± 0.025 with the diet against 0.416 ± 0.020 without, three seeds each; the thesis's own checkpoint reproduces at 0.697 against its reported 0.677. On the identical images and statistic the ten human observers give 0.958 ± 0.019 (range 0.919–0.987) and the CNNs of the Geirhos et al. (2018a) release 0.429, 0.312 and 0.172, values our pipeline reproduces. The diet-trained magno stream, at 0.878, closes most of the distance from the CNN regime toward the observers; no match is claimed — 0.878 is below the lowest individual human — and integration costs 0.203 of shape bias ($0.878 \rightarrow 0.675$).

The integrated readout lands between its two components rather than at the shape-biased one. Given two oppositely biased components that outcome is close to arithmetically forced, so we do not present it as a prediction that could have failed; what can be measured is how much of the rate the readout controls. Sweeping w moves the readout while holding y_1 and y_2 exactly fixed (Fig. 4a): at 21 values of w, three seeds and both checkpoint policies, cue-conflict shape bias runs from 0.40 to 0.90, $w = 0.5$ reproduces the trained model's per-seed values exactly, and the two streams read out alone sit at 0.41 and 0.78 on the forced two-way estimator (Extended Data Table 10), so the span is not an artefact of the decided-image denominator. Because the classifier is affine and the weights sum to one, the sweep is exact interpolation between two measured endpoints; that says the affine readout adds nothing, not that the span would vanish under a readout that is not affine. Under a rule fixed before the test we refitted a two-layer nonlinear head, and a linear head as a control for the refit itself, on the frozen features at each of the 21 weights (Fig. 4b): the nonlinear head's median span is 0.466 against 0.508 for the frozen affine head (86% of the pre-registered baseline, threshold 80%), and the refitted-linear and nonlinear spans agree within 1%. The span is a property of having two oppositely biased components and any readout over them. Restricted to readouts within 5 points of peak validation accuracy it narrows to 0.283 (0.421 to 0.760), so the extreme rates near $w = 1$ are bought with accuracy.

Fig. 4c places every model on one pair of axes. The two streams read out alone bracket the integrated readout (0.903 and 0.400 around 0.655), and the single-stream controls trained with the identical diet lie between (0.567 and 0.626), inside the 0.059 that capacity alone moves the rate; we therefore attribute the ordering among these three models to capacity and claim no shape-bias advantage for the split. The large number on the ladder is the integration cost, 0.248 from the best single component to the integrated readout (Extended Data Table 5).

Nothing here fits w to humans. We tested whether it could be: fitting w to each of the ten observers individually on a 0.001 grid, no observer's rate is reachable in any run (0 of 90 observer-runs; the closest approach misses the lowest observer by 0.027), and a weight fitted to an observer's trial-by-trial choices does not cross-validate (held-out agreement +0.026 against +0.023 for an arbitrary weight, beating it in 45 of 90). The rate is a free parameter of the readout, and nothing in this family of readouts selects the human operating point.

Cortex and behaviour prefer different readout weights. The sweep of Fig. 4a can be run on cortex, because the readout's input $z(w) = w \cdot y_1 + (1 - w) \cdot y_2$ is a feature vector like any other: for each w it was the design of a ridge encoding model of every voxel in NSD's general visual mask, with the

split and folds of the banded model (Methods 4.5), and the w that maximises an ROI's median r^2 is the weight that region prefers (Extended Data Fig. 5; Extended Data Table 12). Under the diet that weight descends the hierarchy — 0.58 in V1, 0.50 in V2, 0.40 in V3, 0.35 in hV4 and 0.33–0.35 in every extrastriate group, ventral, lateral and dorsal alike — with the extrastriate mean below V1 in 8/8 subjects ($\Delta = -0.25$, exact $p = .0078$). The weight that best reproduces the ten observers' trial-by-trial choices is 0.67 (range 0.49–0.89), and the two sets of optima do not overlap (extrastriate 0.27–0.40; Mann–Whitney $p = .0004$). On the axis of the behavioural sweep the two demands sit on opposite sides of the rate: the weight extrastriate cortex prefers gives a cue-conflict shape bias of 0.56, the weight behaviour prefers gives 0.78, and the human rate lies beyond either. Without the diet the gradient reverses (0.65 in V1 to 0.75–1.00 beyond hV4). The curves are shallow — at $w = 0.5$ every ROI is within 3% of its best r^2 — so this is an ordering of optima rather than a large loss, and it is the dissociation of Fig. 3 stated in the readout's own currency: the mixture that best describes extrastriate cortex is parvo-heavy and barely shape-biased, the mixture that best describes behaviour is magno-heavy, and both read out the same two representations.

Without a stream split the readout still moves the rate, and cortex still loads on the texture-leaning units. The span of Fig. 4a is given by construction in a network built around two streams, so we tested networks with none, under a plan committed before the analysis (Methods 4.8). In any frozen network the penultimate units can be split at the median into a shape-leaning and a texture-leaning half by how much of their variance over the cue-conflict images lies between shape rather than texture categories; a gain of 2w on the first half and $2(1 - w)$ on the second, read through the network's own untouched head, is the analogue of the mixing weight, with $w = 0.5$ the trained network (Fig. 6). On the two-stream network's concatenated streams the factorisation recovers them (99% of the shape-leaning half is stream 1) and the sweep reproduces Fig. 4a. In nine networks with no split — our single-trunk controls with and without the diet and six standard ImageNet networks — the rate rises monotonically with w in every network and seed ($\rho \geq 0.71$), by 0.09–0.21 on held-out images, a third to half of the two-stream span: only 0–17% of a single trunk's units are shape-leaning at all, and refitted heads on the lossy sum of the halves give 0.05–0.13 at unchanged accuracy, so the pre-stated magnitude thresholds were met in no single-trunk network and the direction in all (Fig. 6a; Extended Data Table 13). The spans are nonetheless as large as the gaps the field ranks, and reorder the ranking at almost no accuracy cost: ResNet-50 moves from 0.22 past ResNet-152 (0.29) and ResNet-152 past ViT-B/16 (0.40), with ImageNet top-1 within 1 point over the whole sweep for four of the six networks and within 5 points for w in 0.15–0.90 for the ResNets (Fig. 6b). Cortex generalises in part. With the two halves of the no-diet single trunk as encoding bands, the texture-leaning half carries every extrastriate group relative to V1 in 8/8 subjects ($\Delta = -0.32$, exact $p = .0078$), with no dorsal–ventral difference and no descent within V1–hV4, and the cortex-optimal weight of the lossy sum falls from 0.80 in V1 to 0.37–0.48 beyond hV4 (7/8, $p = .023$) while the behavioural rate rises toward $w = 1$ (Fig. 6c, d; Extended Data Table 14). The diet-trained single ResNet-34 shows the same V1-relative contrast (7/8, $p = .016$), a descent already within V1–hV4 (7/8, $p = .016$) and ventral cortex as its most texture-carried region (8/8), but its lossy-sum optimum sits near 0.55 in every region. In ImageNet ResNet-50, whose units are almost all texture-leaning, that half carries V1 through hV4 and ventral cortex alike (index -0.32 to -0.50) while lateral and dorsal cortex are balanced, so the V1-relative prediction fails there (extrastriate − V1 = +0.24, $p = .039$). Unique variances in every single trunk are an order of magnitude smaller than with the split, because the halves of one trunk share most of their variance. The architecture is therefore not what makes the rate a readout property, and in the networks trained on our images it is not what makes extrastriate cortex load on the texture-biased component; it is what makes the components separable enough for one weight to move the rate across most of the range models occupy, and for cortex to be carried almost entirely by one of them.

This is the mechanism we propose for the paradox: a system holding a shape-biased and a texture-biased component, whose behaviour reads out from both, will look shape-biased behaviourally and texture-biased wherever the texture-biased component dominates — and in this model of the subcortical split, the component that dominates the ventral end is the texture-biased one. It is an existence proof with a cortically grounded demonstration, not a measurement of the human readout.

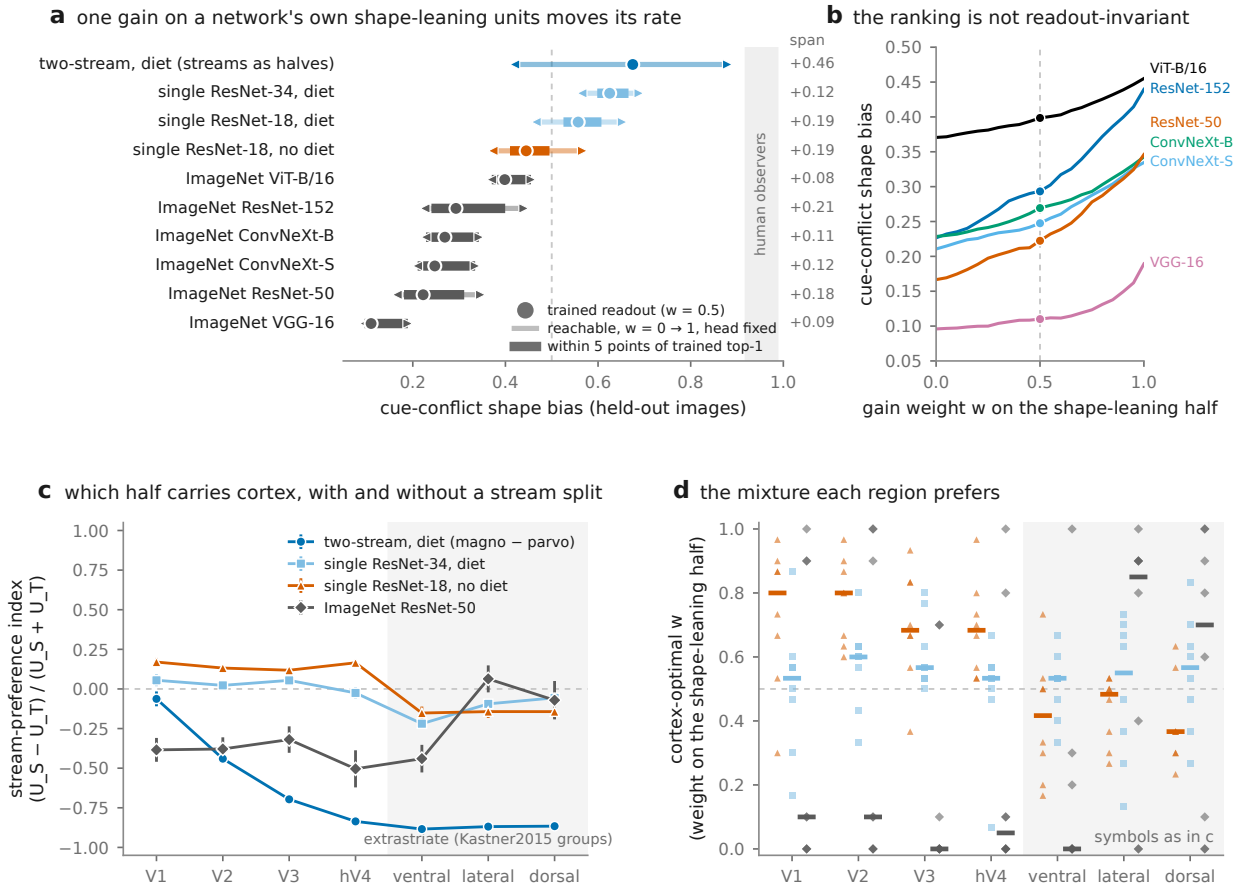


Figure 6 | The readout account beyond the two-stream architecture.

Every network here ends in logits = W·y + b for some penultimate feature vector y. Each unit of y is scored on the 1,200 cue-conflict images by the fraction of its variance lying between the 16 shape categories minus the fraction lying between the 16 texture categories (one-way η^2); the median split gives a shape-leaning half S and a texture-leaning half T. Nothing is retrained.

(**a**) One gain on a network's own shape-leaning units moves its rate. logits(w) = W·(g(w) ⊙ y) + b with g = 2w on S and 2(1 − w) on T, the trained head untouched; w = 0.5 is the trained network (dot). Thin bars: the rate reachable as w runs from 0 to 1, split defined on one stratified half of the images and scored on the other (10 random splits, both directions, mean). Thick segments: the part of the sweep within 5 points of the trained top-1 on held-out natural images (MiniEcoset validation for our networks, a stratified 10-per-class ILSVRC-2012 validation subset for the ImageNet networks). Our networks by the paper's primary estimator, the ImageNet networks by the Geirhos 16-class rule (Methods 4.3). The number at right is the span. In the two-stream network the factorisation recovers the streams — 99% of the shape-leaning half is stream 1 — and the sweep reproduces Fig. 4a (0.42 → 0.88). In the nine networks with no split the rate still rises monotonically with w in every case (Spearman $\rho \geq 0.71$ in every seed and network) but by 0.09–0.21, a third to half of the two-stream span; only 0–17% of a single trunk's units are shape-leaning at all, most in the diet-trained trunks. The sweep is nearly free in accuracy: for ViT-B/16, the two ConvNeXts and VGG-16 top-1 stays within 1 point of the trained value over the whole sweep, and for ResNet-50 and ResNet-152 within 5 points for w between 0.15 and 0.90. Grey band, the ten human observers (0.92–0.99).

(**b**) The same curves for the six ImageNet networks. Dots are the trained rates; the field's ranking is not invariant to the readout: ResNet-50 at w = 1 (0.35) passes ResNet-152's trained rate (0.29), ResNet-152 at w = 1 (0.44) passes ViT-B/16's (0.40), and ConvNeXt-S passes ConvNeXt-B.

(**c**) Which half carries cortex. For the single trunks the two halves are the two bands of a banded ridge model of every NSD voxel (eight subjects, Methods 4.5); the index (U_S − U_T)/(U_S + U_T) is the median over reliable voxels per region, mean ± s.e.m. over subjects (seeds averaged per subject). The two-stream diet model's magno − parvo profile (Fig. 3, Extended Data Fig. 4) is drawn on the same axis. In the no-diet

single trunk the texture-leaning half carries every extrastriate group relative to V1 (8/8 subjects, $\Delta = -0.32$, exact p = .0078) with no descent within V1–hV4; the diet-trained single ResNet-34 shows the V1-relative contrast (7/8, p = .016), a descent already within V1–hV4 (7/8, p = .016) and ventral cortex as its most texture-carried region (dorsal − ventral +0.16, 8/8); in ImageNet ResNet-50, whose units are almost all texture-leaning, that half carries V1 through hV4 and ventral cortex alike (−0.32 to −0.50) while lateral and dorsal cortex are balanced, so the V1-relative contrast reverses (+0.24, p = .039). Unique variances in the single trunks are 0.006–0.04 against 0.09–0.25 with the split.

(**d**) The mixture each region prefers. The lossy sum z(w) = w·y_S + (1 − w)·y_T (halves paired by rank) is the design of a ridge model at 11 values of w (0, 0.1, ..., 1), following Extended Data Fig. 5; points are subjects, bars medians, symbols as in (c). In the no-diet trunk the optimum falls from 0.80 in V1 to 0.37–0.48 beyond hV4 (7/8, p = .023); in the diet ResNet-34 the optimum sits near 0.55 in every region (the curves stay within 5% of their maximum over w in all three networks, as in Extended Data Fig. 5); in ResNet-50, V1–hV4 and ventral cortex prefer the texture-leaning half (0.0–0.1) while lateral and dorsal cortex prefer the shape-leaning one (0.7–0.85). Prediction P5 therefore holds in one of the three networks.

Pre-specification and predictions P1–P5: `docs/PLAN_GENERALITY_2026-09-14.md`; per-network values: Extended Data Tables 13 and 14.

2.5 Properties that do not follow from shape bias

If shape bias is a property of a readout over components rather than of a representation, quantities the field treats as travelling with it need not. Over identical inputs, three do not (Fig. 5).

Robustness. The standard reading of Geirhos et al. (2018a, 2021) is that shape bias buys out-of-distribution robustness. We pre-registered that prediction and tested it on corruptions applied to the image before the front end, scored in the models' own label space at each seed's best saved checkpoint (Fig. 5a; Extended Data Fig. 1; Extended Data Table 6). The diet-trained model is less robust on four of the five corruption families it was not trained on — uniform noise (relative-accuracy AUC 0.634 against 0.729), high-pass (0.678 against 0.827), false colour (0.854 against 0.938) and rotation (0.574 against 0.615) — with complete seed separation on each, and equivalent within $\pm$0.033 on phase scrambling. The only family on which the diet is more robust is low-pass (0.444 against 0.356), declared trained-in before the run because the diet is a low-pass blur; it is not more robust to contrast reduction (0.481 against 0.549) although its front end includes a contrast-sensitivity cut. Robustness appears exactly where it was trained in and nowhere else, which is the strongest available evidence that the nulls are real rather than a broken measurement. This does not contradict Geirhos et al., whose stylisation route yields shape bias together with robustness; it shows the coupling belongs to that route. It differs from the source of the diet, whose title reports robust, shape-based vision: we reproduce the shape-based half (their released model 0.822, our diet-trained magno stream 0.878) and not the robust half, and Methods 4.6 lists the three implementation differences any of which could carry the divergence.

Agreement with human decisions. Two decision-makers with the same marginal rate can agree on no individual image beyond what those rates force; error consistency — Cohen's κ on trial-by-trial agreement — subtracts that forced agreement (Geirhos et al., 2020), and across the ten observers the human–human ceiling is $\kappa = 0.234$. We pre-registered that the diet-trained model would be more human-consistent than its control. It is not, on either readout, and the sharpest true statement is stronger: our most shape-biased model, the diet-trained magno stream at 0.878, has the lowest error consistency of all seven models compared ($\kappa = 0.0089$, 3.8% of the ceiling, against 7.3% for VGG-16 at shape bias 0.172). Across all 189 readouts of the w sweep, with w chosen post hoc per run to maximise agreement, the best κ never exceeds 20.7% of the ceiling (Fig. 5b; Extended Data Table 7), and equivalence intervals fixed in advance exclude a κ change larger than 0.037 across the 0.71 span of shape bias the readouts cover. The measure does detect a correspondence when one is there — the same machinery recovers 24.1% of the split-half ceiling

on whether an image resolves to either cue at all, against 8.7% on which cue is picked (Fig. 5c) — and that positive control invites a diet reading we do not make, because the crossed front-end 2×2 shows a train/test-matching pattern rather than a weights effect (Fig. 5d).

Representational dimensionality. The participation ratio of each stream's features on 1,000 NSD images per subject gives a magno/parvo ratio whose ordering across the four front ends is reproduced by a learning-free Gabor bank, so it tracks the bandwidth of the carrier a stream is handed. What the diet decides is which side of parity training ends on: every diet-trained seed ends magno-dominant (24/24 subject-seeds in each of three diet arms), including the `parvo_highsf` front end, which starts inverted in 8/8 subjects and which we had pre-registered would stay inverted, while every no-diet seed collapses to parity (Extended Data Fig. 2). No level is claimed, because the within-arm seed range exceeds the gaps between arms. The same case separates dimensionality from shape bias: `parvo_highsf`'s second stream becomes the more dimensional stream while remaining texture-biased (probe −0.266).

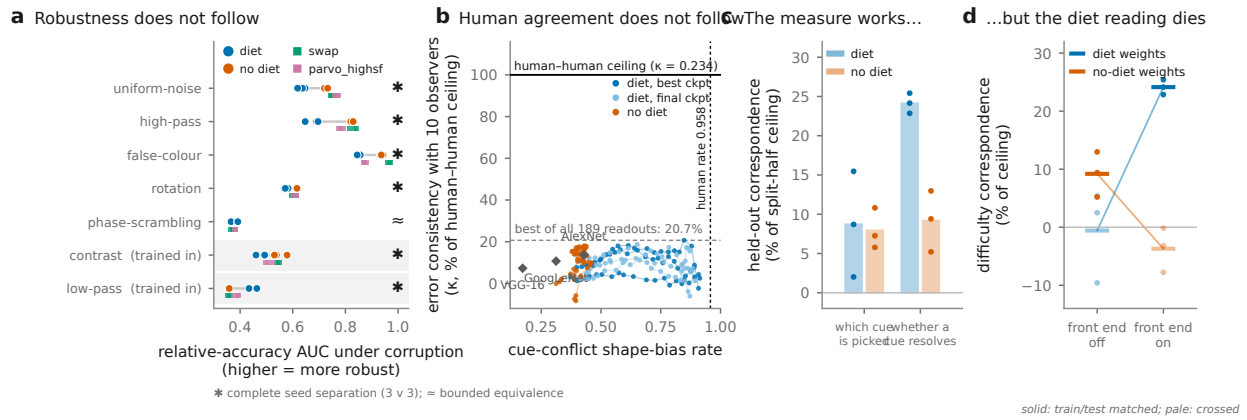


Figure 5 | Three properties the field treats as travelling with shape bias do not follow from it.

(**a**) Out-of-distribution robustness. Corruptions are applied to the image before the front end and scored in the models' own 112-class label space at each seed's best saved checkpoint; plotted is the area under the curve of accuracy relative to each seed's own clean accuracy, one marker per seed, for the diet (blue) and no-diet (vermillion) arms, with the swap and `parvo_highsf` controls as small squares. $*$ marks complete seed separation (the exact 3 v 3 Mann–Whitney floor, $p = 0.100$); $\approx$ marks the one family where the arms are equivalent to within ± 0.033. The diet-trained model is less robust on four of the five corruption families it was not trained on and more robust on exactly one family, low-pass, which the diet installs. Shaded rows are the two families the diet has effectively seen; it is not more robust to contrast reduction even though its front end includes a contrast-sensitivity cut. Full degradation curves: Extended Data Fig. 1.

(**b**) Trial-level agreement with the same ten human observers (error consistency, Cohen's κ, as a percentage of the human–human ceiling of $\kappa = 0.234$ over 45 observer pairs) against cue-conflict shape-bias rate, for all 189 readouts of the w sweep (9 runs × 21 weights) and the three published CNNs (Geirhos et al. 2020). The rate spans 3×; agreement never exceeds 20.7% of the ceiling even with w chosen post hoc, per run, to maximise it. No regression line is drawn, because the x-position of every point was set by a free parameter.

(**c**) Positive control. On the same 1,200 images and the same 10 observers, with shape- and texture-category effects removed and w chosen on half the images and scored on the held-out half, the same machinery recovers 24.1% of the split-half ceiling for whether an image resolves to either cue at all against 8.7% for which cue is picked (diet arm; three seeds as dots, bar = mean). Each target is expressed against its own split-half ceiling.

(**d**) The crossed front-end 2×2 that withdraws the diet reading of (c). Difficulty correspondence for diet weights and no-diet weights evaluated with the developmental front end on and off. Both train/test-matched cells are positive (24.1%, 9.2%) and both crossed cells are not (−0.6%, −3.7%), so the diet arm's advantage in (c) is a train/test-matching pattern, not an effect of the learned weights.

3. Discussion

The two questions. The thesis asked whether magnocellular- and parvocellular-driven streams inside one network could diverge while the network behaves shape-biased, and whether such a system bears on why the ventral stream can be texture-like while behaviour is not. To the first the answer is yes, replicated across seeds and with the mechanism located: divergence requires the developmental transform on an achromatic carrier, the shape-biased component is a property of the input that the diet protects from being unlearned, and the texture-biased component is built by learning. To the second the answer is an existence proof the thesis could not reach with its instruments: fitted to human cortex, it is the texture-biased stream whose unique variance carries the extrastriate end of the hierarchy — ventral, lateral and dorsal alike — while behaviour, reading out from both, is shape-biased. Read as competing claims about a single representation, shape-biased behaviour and texture-like ventral representations cannot both be right; read as claims

about a readout over components, they are not in competition.

What the readout account licenses. A model matching the human shape-bias rate is routinely described as having human-like representations. This system shows that inference fails twice over. The same network holds components at +0.819 and −0.518 and behaves at 0.675, and one scalar over frozen representations moves the rate across most of the range the field's models occupy, so the rate does not identify a representation; and across models spanning a five-fold range of shape bias, trial-by-trial agreement with ten observers stays between 7% and 18% of the human–human ceiling, our most shape-biased stream lowest of all, so the rate does not identify a decision process either. Hermann et al. (2020) and Islam et al. (2021) established that a representation can hold more shape than its behaviour reveals; what is new here is the form of the dissociation — both representations held fixed, the rate moved by the readout alone, a refitted nonlinear head preserving the span — and the cortical grounding, which neither prior study has: the same mixing weight, fitted to cortex, prefers the parvo-heavy end in every region beyond V1, and fitted to human choices prefers the magno-heavy end (Extended Data Fig. 5). Nor does the account need the split: in ten frozen networks with none, one gain on the network's own shape-leaning units moves the rate and reorders the field's ranking at no accuracy cost, and in our single trunks it is again the texture-leaning units that carry extrastriate cortex relative to V1 (Fig. 6); the split amplifies the dissociation, it does not create it. The methodological consequence is direct: shape bias should be reported as the readout statistic it is, alongside error consistency where human trial data exist, and should not on its own license a claim about representations.

What a developmental constraint does. The intuitive account of developmental low-pass input is informational: coarse input protects the learner from texture, so it learns shape. Our results do not support it. The transform removes bandwidth rather than adding structure, and the shape bias it produces is present before training. What the diet decides is what learning does not destroy: without it, training drives the luminance stream from +0.42 to −0.23; with it, +0.80 is still +0.81 after 150 epochs, and a second instrument shows the same signature, the side of parity replicating rather than a level. This is a constraint on the solution learning converges to, not a better input representation — consistent with the thesis's observation that the transformed magno stream becomes less predictive of the category as the parvo stream becomes more so.

What comes apart, and what the diet costs. Shape bias, robustness, dimensionality and human-likeness are discussed as a package. Over identical inputs this system separates them three times, and a readout account says why it should: none of the three is a property of a readout. The diet is also expensive — less accurate (0.556 against 0.701), less stable late in training, less robust on four of five untrained-on corruptions, a worse predictor of BOLD. Nothing here recommends a developmental diet as a training method; its interest is as a model of development, and the thesis's central mechanistic reading — that divergence and shape bias are bought by degrading the magno stream's evidence — survives in every instrument.

Limitations. The load-bearing cortical result rests on eight subjects from one dataset; its magnitude is read as sign and ordering only, and it is the survivor of seven regional hypotheses, disclosed above. Beyond hV4 the index saturates near its bound in every atlas, so the extension establishes that the effect is hierarchy-wide but cannot order extrastriate regions against one another. The mapping between our chromatic stream and texture-selective cortex is by explanatory role, not content: nothing here asserts that the ventral stream is chromatic, the prediction that the chromatic stream should win in colour-biased cortex fails (pooled VO +0.005, bounded within $\pm$0.017), and substituting the koniocellular axis makes it worse (Supplementary Note S2). The `parvo_highsf` condition varies architecture alongside the front end. The diet is a fixed 48-month front end rather than a trajectory, though a schedule-trained checkpoint behaves like it on every instrument we tried. Whether a two-stream architecture specialises on its own is untestable here,

because the trunks are bit-identical at initialisation. Human comparisons are against ten published observers and of aggregate rate; the readout weight is a hyperparameter that nothing selects. The single-trunk factorisation is a median split of units by one cue-conflict score, and its spans are lower bounds on what a readout can reach, not measurements of the best one. Of its five pre-stated predictions, the two behavioural thresholds were met by direction and not by magnitude, the stream-recovery check was met, the cortical contrast held in our two trunks and reversed in ImageNet ResNet-50, whose units are almost all texture-leaning, and the mixing-weight ordering held in one network of three (Extended Data Table 14). Two of the four pre-registered predictions failed and are reported as failures; claims deliberately not made are listed in Supplementary Note S1.

4. Methods

4.1 Front end

RGB images are converted to the DKL cone-opponent space through RGB→XYZ (D65)→LMS (Hunt–Pointer–Estévez)→DKL at 256 × 256 px, each channel normalised to $[0, 1]$ with the training set's global range and quantised to 8 bits exactly as the training data were stored (Derrington et al., 1984). Stream 1 receives the luminance channel; under the diet it passes through an isotropic Gaussian blur ($\sigma = 4$ px, kernel 33, reflect padding) and the scale-free contrast-sensitivity cut of Lu et al. (2026) evaluated at a fixed age of 48 months (`cs_logspan_start` $= 5 \times 10^{-3}$, decomposition mode "keep"). Stream 2 receives the L−M channel untransformed. The four conditions are: `dkl_bcs` (diet: luminance + blur + CS | L−M); `dkl_base` (no diet: luminance | L−M); `dkl_bcs_swap` (luminance | L−M + blur + CS); `dkl_bcs_parvo_highsf` (luminance + blur + CS | L−M stacked with untransformed luminance, two channels). The last differs architecturally in its first-layer fan-in, and that difference is measured rather than argued away (Extended Data Fig. 2; Results 2.5). Evaluation stimuli, including NSD images and cue-conflict images, pass through the identical front end.

4.2 Architecture and training

Two ResNet-18 trunks (He et al., 2016), one per stream, initialised bit-identically, with a joint linear classification head over the weighted sum of their penultimate features (w = 0.5; cross-stream mixing off). MiniEcoset at 256 px, 112 classes (Thorat & Kietzmann, 2024). 150 epochs, batch 256, Adam (lr 1×10^{-4}, $\beta_1 = 0.9$; Kingma & Ba, 2017), cross-entropy, with the augmentation set of Lu et al. (2026) applied on GPU (horizontal flip, rotation $\pm 15°$, colour jitter, random equalise, perspective, sharpness; random grayscale for RGB inputs only). Three seeds per condition; weights saved every 25 epochs. Each run used one NVIDIA RTX A5000 (24 GB); the 21 training runs behind this paper took 578 GPU-hours in total (19.5–51.4 h each, from the runs' own timing logs). All encoding, statistical and plotting work ran on CPU. Single-stream controls use one ResNet-18 or one ResNet-34 trunk fed the diet's luminance input with the identical schedule. The accuracy-matched control retrained `dkl_base` with per-epoch checkpointing and selected by rule the epoch nearest the diet arm's best-checkpoint accuracy. No hyperparameter search was run for any arm: every condition uses the training recipe of Lu et al. (2026) unchanged, and the only selection is the checkpoint policy stated above.

4.3 Behaviour and probe

Cue-conflict shape bias is computed on the Geirhos et al. (2018a) style-transfer set restricted to the 16 categories present in MiniEcoset and to images whose shape and texture categories differ

(1,200 images). Logits are restricted to the 16 columns; the argmax is scored as a shape or texture response; images decided as neither are excluded from the denominator and their count is reported. Bootstrap 95% confidence intervals are over images. A second, forced two-way estimator keeps every image and scores whether the logit of the image's shape category exceeds that of its texture category; it is reported beside the primary estimator for every readout (Extended Data Table 10) and is computed from the same stored logits. The readout is full = classifier($w \cdot y_1 + (1-w) \cdot y_2$), magno_only = classifier($w \cdot y_1$), parvo_only = classifier($(1-w) \cdot y_2$), all through the same classifier and bias term; the w sweep evaluates full at 21 values of w on stored features. The refitted heads (Linear(512, 112) and Linear(512, 512)→ReLU→Linear(512, 112)) were trained on $w \cdot F_1 + (1-w) \cdot F_2$ of MiniEcoset training features at each w with an identical budget (Adam, lr 10^{-3}, batch 1024, 20 epochs) and the same initialisation seed at every w. Human observers are the ten of the same release (1,280 trials each), scored by the identical rule; our recomputation reproduces the published 0.958 and the published AlexNet, GoogLeNet and VGG-16 values. Error consistency is Cohen's κ (Cohen, 1960) on the binary shape/texture choice over images both parties decided, with expected agreement $p_m \cdot p_h + (1-p_m)(1-p_h)$, reported as a percentage of the human–human κ over all 45 observer pairs (Geirhos et al., 2020). Per-image correspondence uses the observers' per-image shape rate and cue-resolution rate (split-half reliabilities r = .658 and .794 after category removal), the models' shape-minus-texture logit margin, w chosen on half the images and scored on the held-out half, as a percentage of each target's own ceiling. The linear probe reduces each stream's features to 256 PCA components and fits multinomial logistic regression (C = 1.0) under 5-fold stratified cross-validation to shape and texture labels separately; bias is shape accuracy minus texture accuracy on the same folds.

4.4 Effective dimensionality

Participation ratio $(\Sigma\lambda)^2/\Sigma\lambda^2$ of the stream feature matrix on 1,000 NSD images per subject (Gao et al., 2017), computed Gram-side, reported as the magno/parvo ratio with parity at 1.0. Subjects are the replication unit within a seed; seeds are reported separately and never pooled. A learning-free reference is the same ratio computed on a Gabor oriented-energy bank applied to each stream's own transformed input.

4.5 Encoding

NSD (Allen et al., 2022) `betas_fithrf_GLMdenoise_RR` single-trial betas, converted to percent signal change and z-scored per session per voxel, `nsdgeneral` mask, eight subjects. Splits are by 73k image id with the shared-515 images held out. Stream features for each subject's stimuli are reduced by PCA to 1,000 components per stream (fit on training images only) and entered as two bands into a banded ridge regression (himalaya; Dupré la Tour et al., 2022) with 3-fold GroupKFold by image id, per-voxel alphas and per-voxel band weights. Unique variance U_m (U_p) is the drop in cross-validated r^2 when the magno (parvo) band is removed. Noise ceilings are NSD's `ncsnr`; voxels below 0.1 are excluded and any ROI cell with fewer than 20 surviving voxels is reported as missing. ROIs are the `prf-visualrois` labels of the Wang et al. (2015) atlas as distributed with NSD (V1v, V1d, V2v, V2d, V3v, V3d, hV4), with hierarchical rank [1, 1, 2, 2, 3, 3, 4]. The canonical-correlation spectrum between the two bands runs from ≈0 to 0.93, so each band retains directions the other cannot predict and unique variance is identifiable; a recovery simulation with the fitted feature geometry and a known across-ROI split returns the true ordering exactly ($\rho = +1.000$ in 8/8 subjects, both arms), which is why the gradient is reported as sign and ordering rather than as calibrated magnitude. Eccentricity bands are NSD's `prf-eccrois` labels (0–0.5°, 0.5–1°, 1–2°, 2–4°, > 4°); continuous pRF eccentricity, size and R^2 are NSD's population-receptive-field estimates in the same functional space, and the within-subject regressions of Extended Data

Table 9 use z-scored log(eccentricity + 0.5) and log(size) over voxels with pRF $R^2 > 5\%$. The per-voxel predicted shape-bias index is obtained by pushing the cue-conflict set through each fitted encoding model. For Extended Data Fig. 4 and Table 11 the same voxelwise fits were relabelled, without refitting, with the full Kastner2015 atlas as distributed with NSD (adding VO1/2, PHC1/2, LO1/2, TO1/2, V3A/B and IPS0–2; IPS3–5, SPL1 and FEF have no voxels inside `nsdgeneral`), with NSD's `streams` parcellation and with the fLoc face, place, body and word ROIs; regional means are unweighted means of the ROI medians, and region contrasts are tested per subject with the exact Wilcoxon test. For the cortex-optimal readout weight (Extended Data Fig. 5; Extended Data Table 12) the classifier's own input $z(w) = w \cdot y_1 + (1 - w) \cdot y_2$, where y_1 and y_2 are the 512-dimensional stream outputs the classifier mixes, was the design of a single-band ridge model of every `nsdgeneral` voxel for $w \in \{0, 0.05, \ldots, 1\}$, with the same split, grouped folds and alpha grid as the banded model and columns z-scored on the training trials; because $z(w)$ sums the streams dimension by dimension the ridge cannot undo the mixture, so w is a constraint on the design rather than a re-parameterisation of the two-band model. w*_cortex is the w that maximises the median test r^2 over an ROI's reliable voxels (seeds averaged per subject), the 1,024-dimensional concatenation $[y_1, y_2]$ is the ceiling any mixture can reach, and the behaviour-optimal weight is the trial-fit w of Fig. 4 (Methods 4.4) for each of the ten observers. The koniocellular control substitutes S−(L+M) for L−M in stream 2 of the untrained seed-0 network, contrast-matched to L−M per image, with the magno band bit-identical across arms (gate $\max|\Delta r^2_magno| < 10^{-4}$, passed at 0.000).

4.6 Out-of-distribution robustness

Corruptions are applied to RGB before the front end and scored in the native 112-class label space. Five families were declared untrained-on before the run (uniform noise, high-pass, phase scrambling, rotation, false colour) and two as trained-in positive controls (low-pass, contrast). Reported as the area under the relative-accuracy curve across severities, at each seed's best saved checkpoint. Three implementation differences from Lu et al. (2026) are declared: our front end is evaluated at a fixed 48 months rather than scheduled (the one we can rule out, since a schedule-trained arm behaves like the static diet); our diet is a transform on one stream's input of a two-stream network rather than an augmentation on a single trunk; and we train on MiniEcoset and score corruptions in that label space, applied before the front end.

4.7 Statistics

Within-subject comparisons use the paired exact Wilcoxon signed-rank test, n = 8 (exact two-sided floor p = .0078); seed-level comparisons use the exact Mann–Whitney U, 3 v 3 (floor p = 0.100, reached only on complete separation), and seeds are never pooled with subjects. Every cortical contrast is computed both on the raw per-voxel difference U_m − U_p and on the normalised index (U_m − U_p)/(U_m + U_p), each summarised by its median over reliable voxels per ROI, and a claim is made only where the two agree; disagreements are reported as unresolved. Equivalence uses two one-sided tests with bounds fixed before any interval was computed (Schuirmann, 1987; Lakens, 2017). No multiple-comparison correction is claimed: with 28 p-values and an exact floor of .0078, Holm–Bonferroni over any family larger than six cannot yield a rejection, so claims rest on complete separation, agreement of both statistics, replication across seeds and effect size (dz = −6.60 for the diet gradient against −1.09 without the diet). Behavioural and robustness results use each seed's best saved checkpoint unless stated; cortical fits use the final checkpoint. Four predictions — the front-end 2×2, the inverted-front-end prediction, the corruption families and the error-consistency hypothesis — were written into a version-controlled plan and committed before the corresponding analysis was run (internal pre-specification, not registry-based); two

failed and are reported as failures.

4.8 Readout re-weighting in networks without a stream split

Pre-specified in `docs/PLAN_GENERALITY_2026-09-14.md`, committed before the analysis was run, with five directional predictions whose outcomes are all reported (Results 2.4; Extended Data Tables 13 and 14). For a frozen network with final layer logits = W·y + b, each unit j of y receives a shape-leaning score d_j = η^2_shape(j) − η^2_texture(j), where η^2 is the one-way fraction of the unit's variance over the 1,200 cue-conflict images lying between the 16 shape categories or the 16 texture categories; the median split of d gives the shape-leaning half S and the texture-leaning half T (equal-sized, as the two streams are). Fixed-head gain sweep: logits(w) = W·(g(w) ⊙ y) + b with g_j = 2w for j ∈ S and 2(1 − w) for j ∈ T, w ∈ {0, 0.05, ..., 1}, so that w = 0.5 is the trained network; the split is defined on one half of the images, stratified by shape × texture pair, and the rate scored on the other, over 10 random splits in both directions. Our networks are scored by the estimators of Methods 4.3; torchvision ImageNet networks (ResNet-50, ResNet-152, VGG-16, ViT-B/16, ConvNeXt-S/B, IMAGENET1K_V1 weights, model-vs-human preprocessing) by the Geirhos 16-class rule, mean softmax probability over each category's ImageNet classes and argmax over the 16, which decides every image. Accuracy under the sweep is top-1 with the same gained features on held-out natural images: the MiniEcoset validation split for our networks and a stratified 10-per-class subset of the ILSVRC-2012 validation set for the ImageNet networks. Because a refitted linear head can undo a per-unit gain, the refit test mixes the halves lossily as the two-stream model does, z(w) = w·y_S + (1 − w)·y_T with the halves paired by rank of d, and refits the linear and two-layer heads of Methods 4.3 at each w. For cortex, S and T (split on all 1,200 images) are the two bands of the banded ridge model of Methods 4.5 on the penultimate units of the single-trunk networks, giving U_S, U_T and the index (U_S − U_T)/(U_S + U_T) per voxel; the cortex-optimal w of the lossy sum follows the protocol of Extended Data Fig. 5 on an 11-point grid (0, 0.1, ..., 1). The `rgb_tsn` network — two trunks that receive the identical three-channel image and remain bit-identical — is one ResNet-18 on the untransformed image and is used as the no-diet single trunk. All of our networks enter this analysis at their final checkpoints, the policy of the cortical fits (Methods 4.7).

Data availability

The Natural Scenes Dataset (Allen et al., 2022) is available at naturalscenesdataset.org under its own data-use agreement, which requires registration; we used the `betas_fithrf_GLMdenoise_RR` (b3) preparation, the `nsdgeneral` mask, the `prf-visualrois` and `prf-eccrois` labels and the pRF estimates as distributed with NSD. The Geirhos cue-conflict stimuli and the ten observers' trial-level responses are distributed at github.com/rgeirhos/texture-vs-shape (directory `raw-data/style-transfer-512-nomask-experiment/`) under CC BY 4.0; the same experiment is mirrored at github.com/bethgelab/model-vs-human, and the two mirrors were parsed by two independently written scripts that agree to four decimals on the human reference. MiniEcoset is at osf.io/msna2 (Thorat & Kietzmann, 2024). Derived result tables and fitted encoding-model summaries sufficient to reproduce every figure and Extended Data table are in the code repository, and will be deposited with a DOI on acceptance.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article; a draft is `docs/REPORTING_SUMMARY.md` in the repository.

Code availability

Training, front-end, encoding, behavioural and analysis code is at github.com/LuZeking/Magno-Parvo-Konio-Net (branch `mpk-rewrite`, commit as stamped on this manuscript's title page), under the BSD-3-Clause licence, with a pinned environment manifest (`requirements.txt`, Python 3.11) and the committed result files behind every figure; the figures of this paper regenerate with the scripts in `tmp_analysis/figs_v2/` and the manuscript's numbers are locked by `tmp_analysis/verify_new_claims.py`. The per-subject encoding outputs that Fig. 3 reads are large and are deposited alongside the derived tables. A private link is available to referees on request; a DOI-archived release accompanies acceptance.

Ethics and human participants

This study collected no new human data. Both human datasets are existing, de-identified public releases analysed under their own terms. NSD was collected under the approval of the University of Minnesota institutional review board with written informed consent from all eight participants (Allen et al., 2022); the behavioural data were collected under the approval described in Geirhos et al. (2018a). Secondary analysis of two de-identified public datasets required no additional approval under our institution's policy, and no participant-identifying information was accessed.

Sex and demographics. Participant sex was not under our control and is not a design factor; no analysis is disaggregated by sex, and at eight participants and ten observers none could be, because the exact tests used already sit at their resolution floor on the full samples. We do not restate sex or age: the Geirhos per-observer files carry only trial-level fields, so no per-observer demographic exists in the data we analysed, and for NSD we have not verified values against the released subject-information file and therefore point to the NSD release rather than transcribe from prose.

Constraints on generality. The neuroimaging sample is eight adults with normal or corrected-to-normal vision in the United States; the behavioural reference is ten adults tested in a German university laboratory. Both are small, Western, convenience-recruited adult samples. Claims about "human observers" should be read as claims about these samples; nothing here establishes cross-cultural, developmental or clinical generality, and the developmental motivation of the model's front end is a motivation for the model, not a claim about any developmental population we measured.

Author contributions

[To be completed at submission per Nature-portfolio policy: conceptualisation, model and training code, encoding analyses, behavioural analyses, figures, writing, supervision. The originating thesis (Manocchio, 2026) supplied the architecture, the front-end conditions and the first behavioural and RSA analyses; the seed replications, encoding models, controls and manuscript were developed subsequently.]

Competing interests

The authors declare no competing interests.

Extended Data tables

Extended Data Table 1 | The cortical gradient hV4 − mean(V1v, V1d), per seed and statistic. "normalised" is the per-voxel index (U_m − U_p)/(U_m + U_p), "raw" the per-voxel difference U_m − U_p, each summarised by its median over reliable voxels per ROI. Mean over 8 subjects of the per-subject contrast; "sep." is the number of subjects in which the diet arm is more negative than the no-diet arm; p is the exact paired Wilcoxon (floor .0078). The n = 7 column drops subject 01 (Results 2.3). Source: `results/encoding_cc/`.

statistic	seed	diet	no diet (n = 8)	no diet (n = 7)	sep.	p
normalised	0	−0.833	−0.078	−0.054	8/8	0.0078
	1	−0.785	−0.096	−0.084	8/8	0.0078
	2	−0.745	−0.049	−0.031	8/8	0.0078
raw	0	−0.166	−0.027	−0.021	8/8	0.0078
	1	−0.145	−0.031	−0.030	8/8	0.0078
	2	−0.153	−0.023	−0.021	8/8	0.0078
untrained, diet front end	—	−0.544			8/8 < 0	0.0078
untrained, no-diet front end	—		−0.479		8/8 < 0	0.0078
swap, seed 0 (degenerate: U_p ≈ 0)	0	−0.006				
parvo_highsf, seed 0 (degenerate: U_m ≈ 0)	0	−0.143				
thesis checkpoint (progressive schedule)	—	−0.793				

Extended Data Table 2 | Monotonicity of the profile and the exact permutation null. Spearman ρ between hierarchical rank [1,1,2,2,3,3,4] and stream preference, mean over subjects; the null enumerates all 630 distinct labellings of the seven ROIs (one-sided in the pre-stated direction, floor 1/630 = .0016). Source: `results/hierarchy_permutation_null.json`.

arm	seed	statistic	observed mean ρ	exact p	most extreme of 630	null 2.5th pct
diet	0	norm	-0.964	0.0016	yes	-0.754
diet	0	raw	-0.964	0.0016	yes	-0.743
diet	1	norm	-0.973	0.0016	yes	-0.750
diet	1	raw	-0.973	0.0016	yes	-0.743
diet	2	norm	-0.966	0.0016	yes	-0.748
diet	2	raw	-0.966	0.0016	yes	-0.756
no diet	0	norm	-0.502	0.0286	no	-0.502
no diet	0	raw	-0.571	0.0206	no	-0.559
no diet	1	norm	-0.606	0.0190	no	-0.546
no diet	1	raw	-0.661	0.0190	no	-0.594

arm	seed	statistic	observed mean ρ	exact p	most extreme of 630	null 2.5th pct
no diet	2	norm	-0.580	0.0365	no	-0.616
no diet	2	raw	-0.463	0.0921	no	-0.594

Extended Data Table 3 | In-silico cue conflict through the fitted encoding models, per seed. Median over reliable voxels of the per-voxel predicted shape-bias index for each stream, mean over 8 subjects; the contrast is magno − parvo, paired within subject against the no-diet arm of the same seed. Source: `results/encoding_cc/`.

seed	arm	magno stream	parvo stream	magno − parvo	diet − no diet	sep.	p
0	diet	+0.791	−0.743	+1.534	+1.424	8/8	0.0078
0	no diet	−0.638	−0.748	+0.110			
1	diet	+0.796	−0.770	+1.566	+1.468	8/8	0.0078
1	no diet	−0.601	−0.699	+0.098			
2	diet	+0.799	−0.759	+1.558	+1.474	8/8	0.0078
2	no diet	−0.613	−0.696	+0.083			

Extended Data Table 4 | The carrier × transform 2×2 on the probe and on behaviour. Mean over training seeds with the seed range in brackets; "decided" is the mean number of the 1,200 cue-conflict images the readout classifies as the shape or the texture category (the shape-bias denominator). Cells are read out on stream 2 in three cases and on the diet's magno stream in the fourth. Source: `results/figures/fig1_data.npz` (from `results/shape_texture/`, `results/cue_conflict/`).

carrier	transform	condition, stream	probe bias	cue-conflict shape bias	decided / 1,200	seeds
chromatic (L−M)	none	no diet, parvo	−0.504 [0.035]	0.364 [0.080]	270	3
chromatic (L−M)	yes	swap, parvo	−0.145 [0.043]	0.495 [0.038]	165	3
achromatic (luminance)	none	parvo_highsf, stream 2	−0.266 [0.017]	0.396 [0.012]	584	2
achromatic (luminance)	yes	diet, magno	+0.819 [0.014]	0.878 [0.022]	488	3

Extended Data Table 5 | Single-stream controls trained with the identical diet, and the readout ladder. Best saved checkpoint per seed; mean [range] over three seeds. Accuracy is validation top-1 at the scored checkpoint. Sources: `results/cue_conflict_bestep/`, `results/cue_conflict_single34_bestep/`, `results/cue_conflict_single_bestep/`, run histories, `results/param_matched/`.

model / readout	parameters	shape bias	decided / 1,200	accuracy
diet, magno stream alone	22,397,936	0.903 [0.888, 0.916]	504	0.556 [0.552, 0.564]
diet, both streams (w = 0.5)	22,397,936	0.655 [0.594, 0.718]	502	0.556 [0.552, 0.564]

model / readout	parameters	shape bias	decided / 1,200	accuracy
diet, parvo stream alone	22,397,936	0.400 [0.372, 0.429]	388	0.556 [0.552, 0.564]
single stream, ResNet-34 (matched)	21,338,992	0.625 [0.604, 0.652]	537	0.623 [0.616, 0.634]
single stream, ResNet-18 (half)	11,230,832	0.567 [0.541, 0.590]	501	0.599 [0.589, 0.608]
no diet, both streams	22,397,936	0.416 [0.394, 0.433]	593	0.701 [0.698, 0.706]

Extended Data Table 6 | Out-of-distribution robustness: relative-accuracy AUC per corruption family and condition. Mean over three seeds [range]; the thesis's schedule-trained checkpoint is a single run. "sep." marks complete seed separation between the diet and no-diet arms (the exact 3 v 3 floor, p = 0.100). Families marked ‡ were declared trained-in before the run. Source: `results/ood_bestep/`.

family	diet	no diet	swap	parvo_highsf	thesis schedule	diet − no diet	sep.
uniform-noise	0.634 [0.619, 0.647]	0.729 [0.720, 0.735]	0.749 [0.744, 0.758]	0.767 [0.758, 0.773]	0.640	−0.095	yes
high-pass	0.678 [0.647, 0.696]	0.827 [0.820, 0.831]	0.831 [0.817, 0.841]	0.783 [0.775, 0.791]	0.680	−0.149	yes
false-colour	0.854 [0.845, 0.860]	0.938 [0.933, 0.943]	0.966 [0.960, 0.971]	0.874 [0.868, 0.880]	0.839	−0.084	yes
rotation	0.574 [0.567, 0.582]	0.615 [0.613, 0.616]	0.605 [0.597, 0.612]	0.611 [0.602, 0.616]	0.572	−0.041	yes
phase-scrambling	0.382 [0.365, 0.391]	0.373 [0.369, 0.377]	0.371 [0.365, 0.375]	0.382 [0.380, 0.383]	0.362	+0.009	no
contrast ‡	0.481 [0.460, 0.493]	0.549 [0.530, 0.578]	0.543 [0.539, 0.550]	0.511 [0.498, 0.526]	0.489	−0.068	yes
low-pass ‡	0.444 [0.434, 0.464]	0.356 [0.353, 0.359]	0.359 [0.353, 0.362]	0.385 [0.375, 0.392]	0.436	+0.089	yes

Extended Data Table 7 | Error consistency (Cohen's κ) with the ten human observers. The trained readout (w = 0.5) and the best readout over the 21 values of w, per run; κ as a percentage of the human–human ceiling (κ = 0.234, 45 pairs). Published CNN values are from Geirhos et al. (2020). Source: `results/sweep_error_consistency/`.

run	shape bias at w = 0.5	κ at w = 0.5	% ceiling	best κ over w	% ceiling	w at best	rate at best
dkl_base_seed1_final	0.421	+0.0309	13.2	+0.0404	17.3	1.00	0.438
dkl_base_seed2_final	0.433	+0.0333	14.3	+0.0421	18.0	0.75	0.429
dkl_base_seed3_final	0.394	+0.0372	15.9	+0.0412	17.6	0.35	0.410

run	shape bias at w = 0.5	κ at w = 0.5	% ceiling	best κ over w	% ceiling	w at best	rate at best
dkl_bcs_seed0_best	0.594	+0.0096	4.1	+0.0178	7.6	1.00	0.890
dkl_bcs_seed0_final	0.692	+0.0234	10.0	+0.0405	17.3	0.90	0.857
dkl_bcs_seed1_best	0.718	+0.0269	11.5	+0.0425	18.2	0.35	0.614
dkl_bcs_seed1_final	0.686	+0.0319	13.7	+0.0437	18.7	0.45	0.651
dkl_bcs_seed2_best	0.652	+0.0372	15.9	+0.0483	20.7	0.85	0.847
dkl_bcs_seed2_final	0.646	+0.0280	12.0	+0.0405	17.3	0.75	0.784
AlexNet (published)	0.429	+0.0320	13.7				
GoogLeNet (published)	0.312	+0.0250	10.7				
VGG-16 (published)	0.172	+0.0170	7.3				

Extended Data Table 8 | The estimator's own bias, and the residual-misfit variants. Synthetic voxels were generated from each arm's fitted 1,000-component feature spaces with the measured per-ROI noise ceilings imposed and the true magno − parvo difference set to zero everywhere; the recovered contrast is the instrument's signed floor. "Displacement" is the real contrast minus the floor, paired per subject. The six variants differ in how the floor is paired (group or per-subject) and whether the residual misfit of the recovery simulation is also removed (before or after rescaling by the recovered slope). Sources: `results/vp_recovery_realnc/diet_floor_paired_2026-09-07.txt`, `floor_pairing_2026-09-07.txt`, `prediction_scored_2026-09-07.txt`.

quantity	diet	no diet
floor, normalised contrast (true = 0)	+0.0375 (8/8 > 0, p = .0078)	+0.0511 (8/8 > 0, p = .0078)
floor, raw contrast	−0.0021 (p = .15)	+0.0009 (p = .84)
real contrast, normalised (seed 0)	−0.8335	−0.0785
displacement real − floor	−0.871 (8/8, p = .0078)	−0.130 (8/8, p = .0078); n = 7: −0.103 (7/7, p = .0156)
recovered slope of estimate vs truth	1.210	1.281
variant: group floor, minimal	−0.720 (8/8, p = .0078)	−0.101 (8/8, p = .0078)
variant: per-subject floor, minimal	−0.720 (8/8, p = .0078)	−0.101 (8/8, p = .0078)
variant: group floor + residual before scaling	−0.626 (8/8, p = .0078)	−0.034 (6/8, p = .055)
variant: per-subject floor + residual before scaling	−0.626 (8/8, p = .0078)	−0.034 (5/8, p = .31)
variant: group floor + residual after scaling	−0.607 (8/8, p = .0078)	−0.015 (4/8, p = 1.00)
variant: per-subject floor + residual after scaling	−0.607 (8/8, p = .0078)	−0.015 (5/8, p = .84)
per-ROI normalised floor, V1v … hV4 (no-diet geometry)		+0.025, +0.025, +0.026, +0.025, +0.027, +0.026, +0.026 (ρ vs rank +0.54, p = .22)

Extended Data Table 9 | Stream preference regressed on pRF eccentricity and pRF size within

subject. Per-voxel stream-preference index regressed on log pRF eccentricity and log pRF size (NSD pRF estimates, voxels with pRF $R^2 > 5\%$, noise ceiling > 0.1); "ecc | size" is the partial slope for eccentricity controlling for size and vice versa; mean over 8 subjects, with the number of subjects on the stated side and the exact Wilcoxon p. Collinearity between the two predictors is modest (mean VIF 1.66). The arm interaction is the paired difference of partial slopes, seed-averaged within arm. Sources: `results/prf_continuous/dissociation_2026-09-08.txt`, `interaction_2026-09-08.txt`, `input_null_2026-09-08.txt`.

model	statistic	ecc alone	size alone	ecc | size	size | ecc
diet, seed 0	normalised	+0.066 (0/8 neg, .0078)	−0.013 (5/8 neg, .31)	+0.119 (0/8 neg, .0078)	−0.086 (8/8 neg, .0078)
diet, seed 1	normalised	+0.065 (0/8, .0078)	−0.029 (7/8, .039)	+0.132 (0/8, .0078)	−0.110 (8/8, .0078)
diet, seed 2	normalised	+0.064 (0/8, .0078)	−0.012 (5/8, .25)	+0.115 (0/8, .0078)	−0.083 (8/8, .0078)
no diet, seed 0	normalised	+0.018 (4/8, .64)	+0.033 (1/8, .039)	−0.004 (4/8, 1.00)	+0.035 (0/8, .0078)
no diet, seed 1	normalised	+0.004 (4/8, .95)	+0.015 (4/8, .38)	−0.009 (6/8, .46)	+0.021 (2/8, .11)
no diet, seed 2	normalised	−0.007 (5/8, .38)	+0.012 (3/8, .31)	−0.022 (6/8, .25)	+0.024 (1/8, .016)
diet, seed 0	raw	+0.103 (1/8, .039)	−0.158 (8/8, .0078)	+0.325 (0/8, .0078)	−0.356 (8/8, .0078)
no diet, seed 0	raw	+0.130 (1/8, .023)	+0.181 (0/8, .0078)	+0.028 (3/8, .84)	+0.163 (0/8, .0078)
Gabor bank, diet front end (input null)	normalised			+0.035 (6/8 pos, .15)	+0.002 (4/8 neg, .95)
untrained network, diet front end	normalised			+0.083 (7/8 pos, .016)	−0.038 (6/8 neg, .039)
arm interaction, diet − no diet (seed-averaged)	normalised			+0.134 (8/8, .0078)	−0.119 (0/8 pos, .0078)
arm interaction, diet − no diet (seed-averaged)	raw			+0.305 (8/8, .0078)	−0.502 (0/8 pos, .0078)

Extended Data Table 10 | Cue-conflict shape bias under the forced two-way estimator. The primary estimator (Methods 4.3) restricts the logits to the 16 categories, takes the argmax, and scores shape against texture over the images decided as one of the two, so images decided as a third category drop out of the denominator ("decided"). The forced estimator keeps all 1,200 images and scores, per image, whether the logit of the shape category exceeds the logit of the texture category. Mean over seeds [range]. Source: the stored per-image logits in `results/cue_conflict*/`.

model / readout	restricted (primary)	decided / 1,200	forced two-way	seeds
diet, magno stream alone (best ckpt)	0.903 [0.888, 0.916]	504	0.775 [0.767, 0.788]	3
diet, both streams w = 0.5 (best ckpt)	0.655 [0.594, 0.718]	502	0.588 [0.574, 0.609]	3
diet, parvo stream alone (best ckpt)	0.400 [0.372, 0.429]	388	0.414 [0.394, 0.428]	3
diet, magno stream alone (final ckpt)	0.878 [0.870, 0.892]	488	0.761 [0.757, 0.764]	3
diet, both streams w = 0.5 (final ckpt)	0.675 [0.646, 0.692]	475	0.608 [0.596, 0.618]	3
diet, parvo stream alone (final ckpt)	0.418 [0.406, 0.438]	353	0.449 [0.411, 0.472]	3
no diet, both streams	0.416 [0.394, 0.433]	593	0.375 [0.357, 0.388]	3
no diet, magno stream alone	0.421 [0.398, 0.434]	602	0.378 [0.352, 0.398]	3
no diet, parvo stream alone	0.364 [0.311, 0.391]	270	0.442 [0.410, 0.473]	3
swap, both streams	0.408 [0.403, 0.412]	586	0.365 [0.362, 0.367]	3
swap, parvo stream alone (L−M + transform)	0.495 [0.478, 0.516]	165	0.511 [0.491, 0.531]	3
parvo_highsf, both streams	0.486 [0.450, 0.516]	618	0.450 [0.438, 0.457]	3
parvo_highsf, stream 2 alone (L−M + luminance)	0.389 [0.374, 0.402]	598	0.346 [0.324, 0.358]	3
single stream, ResNet-34 (matched)	0.625 [0.604, 0.652]	537	0.561 [0.552, 0.573]	3
single stream, ResNet-18 (half)	0.567 [0.541, 0.590]	501	0.523 [0.506, 0.538]	3
no diet, accuracy-matched (epoch 10), magno alone	0.455	532	0.370	1
thesis checkpoint (progressive schedule), both streams	0.697	620	0.636	1
untrained, diet front end, magno alone	0.513	154	0.510	1
untrained, no-diet front end, magno alone	0.568	169	0.508	1

Extended Data Table 11 | Stream preference beyond hV4, on every NSD atlas. Median over reliable voxels of the per-voxel index (Um − Up)/(Um + Up) (noise ceiling > 0.1, Um + Up > 10^{-3}, ≥ 20 voxels per ROI per subject), mean over the eight subjects (n = subjects with enough voxels); trained arms are the mean over training seeds per subject. Positive = magno band, negative = parvo band. Atlases: Wang et al. (2015) Kastner2015 ROIs beyond hV4 (the V1–hV4 rows are Extended Data Table 1 and Fig. 3), NSD's `streams` atlas, and the fLoc category-selective ROIs. Regional contrasts and the exact tests are in Extended Data Fig. 4c and `results/figures_v2/ext_gradient.md`. Source: the stored voxelwise fits in `results/encoding_cc/` and `results/encoding/`, relabelled; no refitting.

atlas	ROI	untrained (diet front end)	no diet (3 seeds)	diet (3 seeds)	swap	parvo + high-SF	thesis ckpt	voxels (median, diet)
Kastner2015	VO1	+0.00 (n=8)	+0.92 (n=8)	−0.87 (n=8)	+0.98 (n=8)	−0.99 (n=8)	−0.94 (n=8)	196
Kastner2015	VO2	+0.12 (n=8)	+0.97 (n=8)	−0.90 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.94 (n=8)	220
Kastner2015	PHC1	+0.41 (n=8)	+0.98 (n=8)	−0.89 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.94 (n=8)	148
Kastner2015	PHC2	+0.46 (n=8)	+0.99 (n=8)	−0.87 (n=8)	+1.00 (n=8)	−0.99 (n=8)	−0.93 (n=8)	150
Kastner2015	LO1	+0.13 (n=8)	+0.92 (n=8)	−0.83 (n=8)	+0.99 (n=8)	−0.98 (n=8)	−0.90 (n=8)	230
Kastner2015	LO2	−0.14 (n=8)	+0.92 (n=8)	−0.90 (n=8)	+0.99 (n=8)	−0.99 (n=8)	−0.94 (n=8)	166
Kastner2015	TO1	+0.25 (n=8)	+0.96 (n=8)	−0.86 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.91 (n=8)	280
Kastner2015	TO2	+0.20 (n=8)	+0.96 (n=8)	−0.89 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.90 (n=8)	36
Kastner2015	V3A	+0.39 (n=8)	+0.91 (n=8)	−0.81 (n=8)	+0.99 (n=8)	−0.98 (n=8)	−0.84 (n=8)	278
Kastner2015	V3B	+0.24 (n=8)	+0.96 (n=8)	−0.87 (n=8)	+1.00 (n=8)	−0.99 (n=8)	−0.94 (n=8)	254
Kastner2015	IPS0	+0.15 (n=8)	+0.98 (n=8)	−0.87 (n=8)	+1.01 (n=8)	−1.00 (n=8)	−0.92 (n=8)	317
Kastner2015	IPS1	−0.06 (n=8)	+0.98 (n=8)	−0.86 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.92 (n=8)	239
Kastner2015	IPS2	+0.02 (n=8)	+1.00 (n=8)	−0.92 (n=8)	+1.01 (n=8)	−1.01 (n=8)	−0.94 (n=8)	47
streams	early	+0.27 (n=8)	+0.86 (n=8)	−0.42 (n=8)	+0.96 (n=8)	−0.92 (n=8)	−0.45 (n=8)	3133
streams	midventral	−0.15 (n=8)	+0.87 (n=8)	−0.84 (n=8)	+0.97 (n=8)	−0.98 (n=8)	−0.86 (n=8)	708
streams	midlateral	−0.04 (n=8)	+0.91 (n=8)	−0.86 (n=8)	+0.98 (n=8)	−0.99 (n=8)	−0.90 (n=8)	571
streams	midparietal	+0.32 (n=8)	+0.94 (n=8)	−0.84 (n=8)	+1.00 (n=8)	−0.99 (n=8)	−0.91 (n=8)	512
streams	ventral	+0.08 (n=8)	+0.96 (n=8)	−0.91 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.95 (n=8)	2724
streams	lateral	+0.25 (n=8)	+0.97 (n=8)	−0.87 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.91 (n=8)	1842
streams	parietal	+0.12 (n=8)	+0.99 (n=8)	−0.87 (n=8)	+1.01 (n=8)	−1.00 (n=8)	−0.93 (n=8)	1082

atlas	ROI	untrained (diet front end)	no diet (3 seeds)	diet (3 seeds)	swap	parvo + high-SF	thesis ckpt	voxels (median, diet)
floc-faces	OFA	−0.07 (n=8)	+0.90 (n=8)	−0.89 (n=8)	+0.98 (n=8)	−1.00 (n=8)	−0.92 (n=8)	367
floc-faces	FFA-1	+0.01 (n=8)	+0.95 (n=8)	−0.91 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.94 (n=8)	399
floc-faces	FFA-2	+0.22 (n=7)	+0.98 (n=7)	−0.92 (n=7)	+0.99 (n=7)	−1.00 (n=7)	−0.95 (n=7)	260
floc-places	OPA	+0.34 (n=8)	+0.98 (n=8)	−0.86 (n=8)	+1.01 (n=8)	−1.00 (n=8)	−0.93 (n=8)	1070
floc-places	PPA	+0.44 (n=8)	+0.99 (n=8)	−0.88 (n=8)	+1.00 (n=8)	−0.99 (n=8)	−0.94 (n=8)	620
floc-bodies	EBA	+0.17 (n=8)	+0.97 (n=8)	−0.88 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.92 (n=8)	2174
floc-bodies	FBA-1	−0.07 (n=6)	+0.96 (n=6)	−0.91 (n=6)	+1.00 (n=6)	−1.00 (n=6)	−0.95 (n=6)	314
floc-bodies	FBA-2	+0.13 (n=8)	+0.98 (n=8)	−0.92 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.94 (n=8)	252
floc-words	OWFA	−0.29 (n=8)	+0.88 (n=8)	−0.85 (n=8)	+0.98 (n=8)	−0.99 (n=8)	−0.88 (n=8)	392
floc-words	VWFA-1	−0.16 (n=8)	+0.96 (n=8)	−0.91 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.96 (n=8)	454
floc-words	VWFA-2	−0.02 (n=5)	+0.95 (n=6)	−0.94 (n=6)	+0.99 (n=6)	−1.00 (n=6)	−0.96 (n=6)	150
floc-words	mfs-words	−0.08 (n=8)	+0.97 (n=8)	−0.93 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.96 (n=8)	121

Extended Data Table 12 | The readout weight cortex prefers against the one behaviour prefers. For each mixing weight w the classifier's own input $z(w) = w \cdot y_1 + (1 - w) \cdot y_2$ (y_1 magno stream output, y_2 parvo stream output, 512-d each) is the design of a ridge encoding model of every nsdgeneral voxel (same split, folds and alpha grid as the banded model; Methods 4.5). w*_cortex is the w that maximises the median test R^2 over reliable voxels of the ROI (mean over seeds per subject; median [range] over 8 subjects). "R^2 at w = 0.5 / max" is the median R^2 at the paper's readout weight relative to the best w; "concat" is the median R^2 of the 1,024-d concatenation, the ceiling for any mixture. "Voxels magno / parvo" is the fraction of the ROI's voxels whose own best w is ≥ 0.75 / ≤ 0.25. The behaviour-optimal weight (trial fit to each of 10 human observers' cue-conflict choices, diet final checkpoints, 3 seeds) is given in the last row for comparison, with the cue-conflict shape-bias rate the diet network produces at each weight.

arm	ROI	w*_cortex median [range]	R^2 at w = 0.5 / max	max R^2 (median)	concat R^2	voxels best w ≥ 0.75	voxels best w ≤ 0.25	rate at w*_cortex	n
diet	V1	0.58 [0.50, 0.70]	0.994	0.1313	0.1402	0.19	0.11	0.72	8
diet	V2	0.50 [0.45, 0.55]	0.999	0.1024	0.1066	0.12	0.20	0.67	8

arm	ROI	w*_cortex median [range]	R^2 at w = 0.5 / max	max R^2 (median)	concat R^2	voxels best w ≥ 0.75	voxels best w ≤ 0.25	rate at w*_cortex	n
diet	V3	0.40 [0.30, 0.50]	0.987	0.0847	0.0864	0.08	0.29	0.61	8
diet	hV4	0.35 [0.20, 0.50]	0.976	0.0945	0.0948	0.05	0.41	0.57	8
diet	ventral (VO, PHC)	0.33 [0.25, 0.35]	0.971	0.1339	0.1325	0.02	0.46	0.56	8
diet	lateral (LO, TO)	0.33 [0.25, 0.45]	0.974	0.1067	0.1060	0.03	0.40	0.56	8
diet	dorsal (V3A/B, IPS0–2)	0.35 [0.25, 0.45]	0.982	0.0683	0.0673	0.05	0.39	0.57	8
no diet	V1	0.65 [0.55, 0.80]	0.987	0.1221	0.1259	0.43	0.09	nan	8
no diet	V2	0.58 [0.40, 0.95]	0.995	0.0968	0.1022	0.36	0.13	nan	8
no diet	V3	0.58 [0.50, 1.00]	0.991	0.0845	0.0890	0.40	0.13	nan	8
no diet	hV4	0.62 [0.40, 0.70]	0.988	0.0965	0.1014	0.40	0.09	nan	8
no diet	ventral (VO, PHC)	0.75 [0.65, 1.00]	0.975	0.1557	0.1535	0.59	0.02	nan	8
no diet	lateral (LO, TO)	0.88 [0.60, 1.00]	0.949	0.1190	0.1214	0.59	0.03	nan	8
no diet	dorsal (V3A/B, IPS0–2)	1.00 [0.75, 1.00]	0.944	0.0805	0.0792	0.61	0.04	nan	8
behaviour (diet)	10 human observers, trial fit	0.67 [0.49, 0.89]	—	—	—	—	—	0.78	10

Extended Data Table 13 | The readout span without a stream split: fixed-head gain sweep on ten frozen networks. Each unit of a network's penultimate layer is scored by the fraction of its variance between shape categories minus that between texture categories on the 1,200 cue-conflict images (one-way η^2); the median split gives a shape-leaning half S and a texture-leaning half T. $\text{logits}(w) = \text{head}(g(w) \odot y)$ with $g = 2w$ on S and $2(1 - w)$ on T, the network's own head untouched; $w = 0.5$ is the trained network. The split is defined on one stratified half of the images and the rate scored on the other (10 random splits, both directions; mean, with the range over splits and seeds in brackets). Rates by the paper's primary estimator (112-class, decided images) for our

networks and the Geirhos 16-class probability aggregation for the ImageNet networks; "forced-estimator span" is the same sweep on the forced two-way estimator. "shape-leaning units" is the fraction of units with η^2_shape > η^2_texture on all images; for the two-stream concatenation the fraction of S that belongs to stream 1 is given in parentheses (P3). "w within 5 pts" is the w-range over which top-1 on held-out natural images (MiniEcoset validation; a stratified 10-per-class ILSVRC-2012 validation subset for the ImageNet networks) stays within 5 points of the trained value. "refit span" is the lossy-sum protocol $z(w) = w \cdot y_S + (1 - w) \cdot y_T$ with a linear and a two-layer head refitted at every w (Methods 4.3 budget), our networks only. Pre-specification: `docs/PLAN_GENERALITY_2026-09-14.md`. Source: `/scratch/byrong/mpk/generality/behaviour/2026-09-14_gen/`.

network	units	trained rate	w = 0	w = 1	span (held-out) [range]	ρ(w, rate)	forced-estimator span	shape-leaning units	w within 5 pts of trained top-1	refit span linear / two-layer	seeds
two-stream, diet, streams concatenated (validation)	1024	0.675	0.420	0.879	0.461 [0.39, 0.57]	1.00	0.307	0.49 (S from stream 1: 0.99)	—	—	3
single ResNet-34, diet	512	0.625	0.565	0.688	0.136 [0.03, 0.24]	0.71	0.102	0.17	0.20–0.78 (top-1 0.626)	0.06 / 0.05	3
single ResNet-18, diet	512	0.557	0.467	0.652	0.194 [0.11, 0.31]	0.97	0.131	0.09	0.28–0.72 (top-1 0.547)	0.09 / 0.06	3
single ResNet-18, no diet (`rgb_tsn`)	512	0.445	0.374	0.566	0.195 [0.11, 0.39]	0.96	0.164	0.05	0.27–0.75 (top-1 0.678)	0.10 / 0.09	3
ImageNet ViT-B/16	768	0.399	0.371	0.455	0.086 [0.07, 0.11]	0.98	0.138	0.01	0.00–1.00 (top-1 0.815)	—	1
ImageNet ResNet-152	2048	0.293	0.227	0.440	0.214 [0.18, 0.25]	1.00	0.196	0.02	0.15–0.90 (top-1 0.785)	—	1
ImageNet ConvNeXt-B	1024	0.269	0.229	0.343	0.115 [0.09, 0.14]	0.99	0.173	0.00	0.00–1.00 (top-1 0.843)	—	1

network	units	trained rate	w = 0	w = 1	span (held-out) [range]	ρ(w, rate)	forced-estimator span	shape-leaning units	w within 5 pts of trained top-1	refit span linear / two-layer	seeds
ImageNet ConvNeXt-S	768	0.248	0.211	0.335	0.124 [0.11, 0.15]	0.99	0.179	0.00	0.00–1.00 (top-1 0.840)	—	1
ImageNet ResNet-50	2048	0.222	0.167	0.346	0.180 [0.14, 0.23]	0.99	0.202	0.02	0.15–0.90 (top-1 0.760)	—	1
ImageNet VGG-16	4096	0.110	0.096	0.189	0.095 [0.07, 0.11]	0.96	0.106	0.01	0.00–1.00 (top-1 0.711)	—	1

Extended Data Table 14 | Cortex without a stream split: the shape- and texture-leaning halves as encoding bands. For each single-trunk network the penultimate units were split as in Extended Data Table 13 (all 1,200 images) and the two halves entered as two bands of a banded ridge model of every `nsdgeneral` voxel (same split, folds and alpha grid as Methods 4.5); U_shape and U_texture are the noise-ceiling-normalised unique variances and the index is (U_shape − U_texture)/(U_shape + U_texture), median over reliable voxels (noise ceiling > 0.1, ≥ 20 voxels), mean over eight subjects (seeds averaged per subject). Regions: V1–hV4 from `prf-visualrois`; ventral (VO1/2, PHC1/2), lateral (LO1/2, TO1/2) and dorsal (V3A/B, IPS0–2) from Kastner2015. The two-stream diet row applies the same computation to the magno (shape-biased) and parvo (texture-biased) bands of Fig. 3. w*_cortex is the mixing weight of the lossy sum z(w) = w·y_S + (1 − w)·y_T that maximises an ROI's median test R^2 (Extended Data Fig. 5 protocol). Exact paired Wilcoxon over subjects. Source: `/scratch/byrong/mpk/generality/cortex/2026-09-14_gen/`.

network	statistic	V1	V2	V3	hV4	ventral	lateral	dorsal	hV4 − V1 (n neg, p)	extrastriate − V1 (n neg, p)	peak-rank, index) V1–hV4
two-stream, diet (magno − parvo)	index	-0.06	-0.44	-0.70	-0.84	-0.88	-0.87	-0.87	—	—	—
	U_shape	0.087	0.045	0.026	0.016	0.014	0.014	0.013			
	U_texture	0.099	0.119	0.145	0.180	0.246	0.190	0.181			
single ResNet-34, diet	index	+0.05	+0.02	+0.05	-0.03	-0.22	-0.09	-0.06	-0.080 (7/8, 0.0156)	-0.178 (7/8, 0.0156)	-0.25
	U_shape	0.032	0.030	0.028	0.025	0.017	0.016	0.020			
	U_texture	0.028	0.028	0.025	0.027	0.027	0.020	0.022			

network	statistic	V1	V2	V3	hV4	ventral	lateral	dorsal	hV4 − V1 (n neg, p)	extrastriate − V1 (n neg, p)	ρ(rank, index) V1–hV4
	w*_cortex (median)	0.53	0.60	0.57	0.53	0.53	0.55	0.57		Δw* extrastriate − V1 +0.01 (4/8 neg, 0.9453)	
single ResNet-18, no diet	index	+0.17	+0.13	+0.12	+0.17	-0.15	-0.14	-0.14	-0.005 (5/8, 0.9453)	-0.316 (8/8, 0.0078)	-0.15
	U_shape	0.038	0.035	0.030	0.031	0.019	0.016	0.018			
	U_texture	0.026	0.026	0.024	0.022	0.025	0.022	0.024			
	w*_cortex (median)	0.80	0.80	0.68	0.68	0.42	0.48	0.37		Δw* extrastriate − V1 -0.32 (7/8 neg, 0.0234)	
ImageNet ResNet-50	index	-0.38	-0.38	-0.32	-0.50	-0.44	+0.06	-0.07	-0.120 (5/8, 0.3828)	+0.235 (1/8, 0.0391)	-0.13
	U_shape	0.008	0.008	0.008	0.006	0.006	0.010	0.009			
	U_texture	0.019	0.018	0.017	0.020	0.015	0.009	0.011			
	w*_cortex (median)	0.10	0.10	0.00	0.05	0.00	0.85	0.70		Δw* extrastriate − V1 +0.07 (3/8 neg, 0.8125)	

References

Allen, E. J., St-Yves, G., Wu, Y., Breedlove, J. L., Prince, J. S., Dowdle, L. T., Nau, M., Caron, B., Pestilli, F., Charest, I., Hutchinson, J. B., Naselaris, T., & Kay, K. (2022). A massive 7T fMRI dataset to bridge cognitive neuroscience and artificial intelligence. *Nature Neuroscience*, 25(1), 116–126. https://doi.org/10.1038/s41593-021-00962-x

Bakhtiari, S., Mineault, P., Lillicrap, T., Pack, C., & Richards, B. (2021). The functional specialization of visual cortex emerges from training parallel pathways with self-supervised predictive learning. *Advances in Neural Information Processing Systems*, 34, 25164–25178.

Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1), 37–46. https://doi.org/10.1177/001316446002000104

Derrington, A. M., & Lennie, P. (1984). Spatial and temporal contrast sensitivities of neurones in lateral geniculate nucleus of macaque. *The Journal of Physiology*, 357(1), 219–240. https://doi.org/10.1113/jphysiol.1984.sp015498

Derrington, A. M., Krauskopf, J., & Lennie, P. (1984). Chromatic mechanisms in lateral geniculate nucleus of macaque. *The Journal of Physiology*, 357(1), 241–265. https://doi.org/10.1113/jphysiol.1984.sp015499

Dupré la Tour, T., Eickenberg, M., Nunez-Elizalde, A. O., & Gallant, J. L. (2022). Feature-space selection with banded ridge regression. *NeuroImage*, 264, 119728. https://doi.org/10.1016/j.neuroimage.2022.119728

Ferrera, V. P., Nealey, T. A., & Maunsell, J. H. R. (1994). Responses in macaque visual area V4 following inactivation of the parvocellular and magnocellular LGN pathways. *Journal of Neuroscience*, 14(4), 2080–2088. https://doi.org/10.1523/JNEUROSCI.14-04-02080.1994

Gao, P., Trautmann, E., Yu, B., Santhanam, G., Ryu, S., Shenoy, K., & Ganguli, S. (2017). A theory of multineuronal dimensionality, dynamics and measurement. *bioRxiv*. https://doi.org/10.1101/214262

Geirhos, R., Rubisch, P., Michaelis, C., Bethge, M., Wichmann, F. A., & Brendel, W. (2018a). ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. *International Conference on Learning Representations* (ICLR 2019). https://openreview.net/forum?id=Bygh9j09KX

Geirhos, R., Meding, K., & Wichmann, F. A. (2020). Beyond accuracy: Quantifying trial-by-trial behaviour of CNNs and humans by measuring error consistency. *Advances in Neural Information Processing Systems*, 33. https://doi.org/10.48550/arXiv.2006.16736

Geirhos, R., Narayanappa, K., Mitzkus, B., Thieringer, T., Bethge, M., Wichmann, F. A., & Brendel, W. (2021). Partial success in closing the gap between human and machine vision. *Advances in Neural Information Processing Systems*, 34, 23885–23899. https://doi.org/10.48550/arXiv.2106.07411

Goodale, M. A., & Milner, A. D. (1992). Separate visual pathways for perception and action. *Trends in Neurosciences*, 15(1), 20–25. https://doi.org/10.1016/0166-2236(92)90344-8

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *IEEE Conference on Computer Vision and Pattern Recognition*, 770–778. https://doi.org/10.1109/CVPR.2016.90

Hermann, K., Chen, T., & Kornblith, S. (2020). The origins and prevalence of texture bias in convolutional neural networks. *Advances in Neural Information Processing Systems*, 33. https://doi.org/10.48550/arXiv.1911.09071

Islam, M. A., Kowal, M., Esser, P., Jia, S., Ommer, B., Derpanis, K. G., & Bruce, N. (2021). Shape or texture: Understanding discriminative features in CNNs. *International Conference on Learning Representations*. https://doi.org/10.48550/arXiv.2101.11604

Jagadeesh, A. V., & Gardner, J. L. (2022). Texture-like representation of objects in human visual cortex. *Proceedings of the National Academy of Sciences*, 119, e2115302119. https://doi.org/10.1073/pnas.2115302119

Jagadeesh, A. V., & Livingstone, M. S. (2024). Texture bias in primate ventral visual cortex. *ICLR 2024 Workshop on Representational Alignment*. https://openreview.net/forum?id=D6YvyvZ7P7

Jang, H., & Tong, F. (2021). Convolutional neural networks trained with a developmental sequence of blurry to clear images reveal core differences between face and object processing. *Journal of Vision*, 21(12), 6. https://doi.org/10.1167/jov.21.12.6

Kaplan, E., & Shapley, R. M. (1982). X and Y cells in the lateral geniculate nucleus of macaque monkeys. *The Journal of Physiology*, 330(1), 125–143. https://doi.org/10.1113/jphysiol.1982.sp014333

Kingma, D. P., & Ba, J. (2017). Adam: A method for stochastic optimization. *arXiv*. https://doi.org/10.48550/arXiv.1412.6980

Komatsu, H., & Goda, N. (2018). Neural mechanisms of material perception: Quest on Shitsukan. *Neuroscience*, 392, 329–347. https://doi.org/10.1016/j.neuroscience.2018.09.001

Kriegeskorte, N., Mur, M., & Bandettini, P. A. (2008). Representational similarity analysis — connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2, 4. https://doi.org/10.3389/neuro.06.004.2008

Lakens, D. (2017). Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4), 355–362. https://doi.org/10.1177/1948550617697177

Livingstone, M. S., & Hubel, D. H. (1987). Psychophysical evidence for separate channels for the perception of form, color, movement, and depth. *Journal of Neuroscience*, 7(11), 3416–3468. https://doi.org/10.1523/JNEUROSCI.07-11-03416.1987

Lu, Z., Thorat, S., Cichy, R. M., & Kietzmann, T. C. (2026). Adopting a human developmental visual diet yields robust and shape-based AI vision. *Nature Machine Intelligence*, 8(5), 735–748. https://doi.org/10.1038/s42256-026-01228-6

Manocchio, B. (2026). *Investigating two-stream artificial neural networks using a developmental visual diet* [Bachelor's thesis, University of Osnabrück, Institute of Cognitive Science; supervised by Z. Lu and T. C. Kietzmann].

Mishkin, M., Ungerleider, L. G., & Macko, K. A. (1983). Object vision and spatial vision: Two cortical pathways. *Trends in Neurosciences*, 6, 414–417. https://doi.org/10.1016/0166-2236(83)90190-X

Nassi, J. J., & Callaway, E. M. (2009). Parallel processing strategies of the primate visual system. *Nature Reviews Neuroscience*, 10(5), 360–372. https://doi.org/10.1038/nrn2619

Okazawa, G., Tajima, S., & Komatsu, H. (2015). Image statistics underlying natural texture selectivity of neurons in macaque V4. *Proceedings of the National Academy of Sciences*, 112(4), E351–E360. https://doi.org/10.1073/pnas.1415146112

Schuirmann, D. J. (1987). A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6), 657–680. https://doi.org/10.1007/BF01068419

Thorat, S., & Kietzmann, T. C. (2024). MiniEcoset. https://doi.org/10.17605/OSF.IO/MSNA2

Wang, L., Mruczek, R. E. B., Arcaro, M. J., & Kastner, S. (2015). Probabilistic maps of visual topography in human cortex. *Cerebral Cortex*, 25(10), 3911–3931. https://doi.org/10.1093/cercor/bhu277

Supplementary Note S1 — claims deliberately not made

- That the two-stream split is what makes shape bias a readout property: single trunks with no split show the same dissociation in kind — a gain moves the rate in every network, and in our trunks extrastriate cortex loads on the texture-leaning half relative to V1 — at a third of the magnitude, with the cortical part failing in ImageNet ResNet-50 (Fig. 6; Results 2.4).
- That the two-stream split causes a shape-bias advantage over a single stream: at matched capacity the arms do not separate at n = 3, and capacity alone moves the rate by a comparable amount (Results 2.4).
- Any dorsal/ventral stream mapping. The thesis's framing "magno→dorsal, parvo→ventral" was tested rather than dropped: the pre-declared dorsal−ventral interaction is +0.0245 on the diet model and +0.0122 (p = .195) without the low-pass front end, so it does not survive its own control. Read directly on the Wang atlas, dorsal (V3A/B, IPS0–2) and ventral (VO1/2, PHC1/2) regions are indistinguishable under both arms (dorsal − ventral +0.015 with the diet, +0.005 without, p > .1; Extended Data Fig. 4c): the diet's magno stream contributes no unique variance anywhere beyond V3.
- hMT+ specificity: at n = 8 the two statistics disagree (p = .195 normalised, .039 raw) and the effect is dead in the blur control.
- A foveal/peripheral headline: the foveal band's sign flips when the blur is removed (−0.057 → +0.051), so it is a front-end artefact.
- That the chromatic stream predicts colour-biased cortex: pooled VO is +0.0045 (p = .46), bounded within ±0.017; hV4's parvo preference is weak (−0.012, 2/8). Substituting the koniocellular axis makes the prediction worse, and worse specifically in the colour set (hV4 −0.029, VO −0.024, each p = .0078, 0/8).
- Where in cortex the diet's shape-bias effect lands: a diet-specific hV4/LO step does not survive the full control set.
- That integration produces human-like shape bias, or that the diet improves neural predictivity: it does neither.
- That the diet makes the model's image-level difficulty profile more human-like: the crossed 2×2 shows train/test matching, not a weights effect.
- That human cortex uses this mechanism: a single-stream account in which texture-like ventral representations feed shape-biased behaviour directly is not excluded by these data.

Why the three regional claims above failed together, and what it implies for the survivor: without the low-pass front end essentially every ROI is magno-preferring, so the blur manufactures apparent regional specificity by handicapping the magno stream until the parvo stream can win somewhere — the same general-factor pathology that defeated the thesis's RSA, reappearing in encoding form. Any regional claim on these models must be shown to survive removal of the blur, and the ventral gradient is reported because it does.

Supplementary Note S2 — the koniocellular control

The thesis omitted the koniocellular pathway for simplicity. Because L−M fails to predict colour-biased cortex, two readings were live: that the chromatic stream does not predict colour cortex, or that L−M is the wrong chromatic axis. Swapping S−(L+M) into stream 2 of the untrained network, contrast-matched to L−M per image, with the luminance stream bit-identical, makes the ventral prediction worse and worse where colour lives (hV4 −0.029, VO −0.024, the pre-declared colour set −0.026, each p = .0078 with 0/8 subjects positive on both statistics; colour set minus the other 17 ROIs −0.014, p = .0078), unchanged when the channel keeps its native 2.05× contrast.

Both cone-opponent axes have now been tested against colour-biased cortex and neither predicts it. These fits are untrained and single-seed, no third stream was added, and no blur was applied to the S channel; whether a three-band model beats a two-band one is a separate question requiring three-way variance partitioning.

Supplementary Note S3 — the cortical gradient against its nuisance axes and the instrument's bias

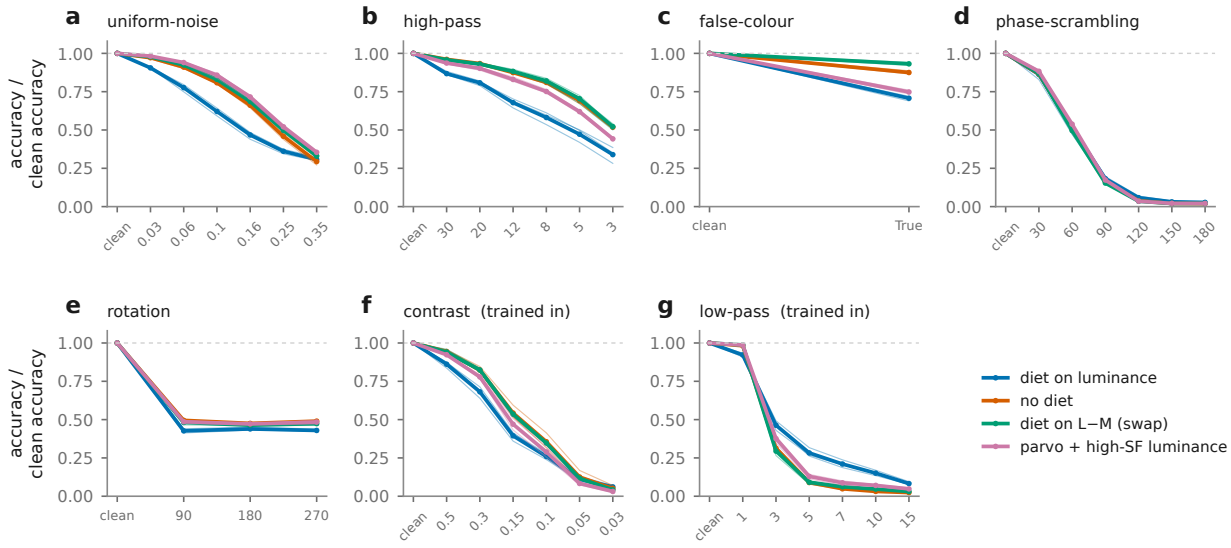
Results 2.3 states the outcome; this note carries the analyses. See Extended Data Fig. 3 and Extended Data Tables 8 and 9.

The two nuisance axes that co-vary with hierarchical rank. Receptive-field size and eccentricity coverage both rise from V1 to hV4, and the diet's magno stream is a low-pass instrument by construction, so a stream-preference gradient that merely tracked receptive-field size would be expected from the front end alone. We tested this three ways (Extended Data Fig. 3; Extended Data Table 9). First, the gradient computed within each of NSD's five eccentricity bands — V1 and hV4 voxels compared only with voxels of the same eccentricity — is negative under the diet in every band (6/7 subjects in the foveal band, 8/8 in the other four) while the no-diet arm stays between -0.10 and $+0.07$ in every band; the diet arm's gradient does grow with eccentricity (-0.26 in the foveal band, -0.93 beyond 4°), so eccentricity modulates it, but the contrast is present at matched eccentricity and is therefore not an eccentricity contrast in disguise. Second, stratifying the same voxels by pRF size gives the same result within size tertiles (-0.71, -0.77 and -0.90 under the diet against -0.08 in each tertile without it). Third, regressing the per-voxel index on log pRF eccentricity and log pRF size jointly within subject (NSD's pRF estimates, voxels with pRF $R^2 > 5\%$; the two predictors are only modestly collinear, VIF 1.66) shows that under the diet, stream preference moves toward parvo with increasing pRF size at fixed eccentricity (partial slope -0.086, 8/8 subjects, $p = .0078$) and toward magno with increasing eccentricity at fixed size ($+0.119$, 8/8, $p = .0078$), while the no-diet arm shows neither the size effect in that direction ($+0.035$, opposite sign) nor any eccentricity effect (-0.004, $p = 1.00$); the arm interaction on the eccentricity partial is $+0.134$ (8/8, $p = .0078$). A learning-free Gabor bank on the diet front end carries no such structure ($+0.035$, $p = .15$), so the dissociation is something training installs, not a property of the input filter. Within the diet arm, then, the ventral gradient and the pRF-size dependence are the same fact seen on two axes — larger receptive fields, further along the hierarchy, are increasingly explained by the texture-biased stream — and neither is present without the diet.

The estimator's own bias points the wrong way. The mean noise ceiling declines from V1v to hV4 ($\rho = -0.89$ against rank), a nuisance falling along the axis of the effect, so we simulated the instrument: synthetic voxels from each arm's fitted feature spaces, the measured noise ceilings imposed, the true magno − parvo difference set to zero everywhere. The recovered contrast is a small, positive floor ($+0.038$ with the diet's feature geometry, $+0.051$ without, 8/8 subjects each), flat across ROIs ($+0.025$ to $+0.027$ per ROI), and indistinguishable from zero on the raw statistic — opposite in sign to the observed effect and without its gradient. Because the floor is a signed bias, the right comparison is a paired displacement: the real contrast minus each subject's own floor is -0.871 with the diet and -0.130 without, each non-zero in every subject, the diet arm's 6.7 times larger (Extended Data Table 8). The floor subtraction relocates the recovery simulation's residual misfit rather than removing it, and we do not correct that residual, because the correction is under-specified: six variants swing the no-diet arm between 4/8 and 6/8 subjects and between $p = .055$ and $p = 1.00$, while the diet arm is non-zero in all eight subjects at $p = .0078$ under all six (Extended Data Table 8). The no-diet arm's non-zero gradient therefore rests on the floor correction alone, and since the simulation is not magnitude-faithful (it recovers the true contrast

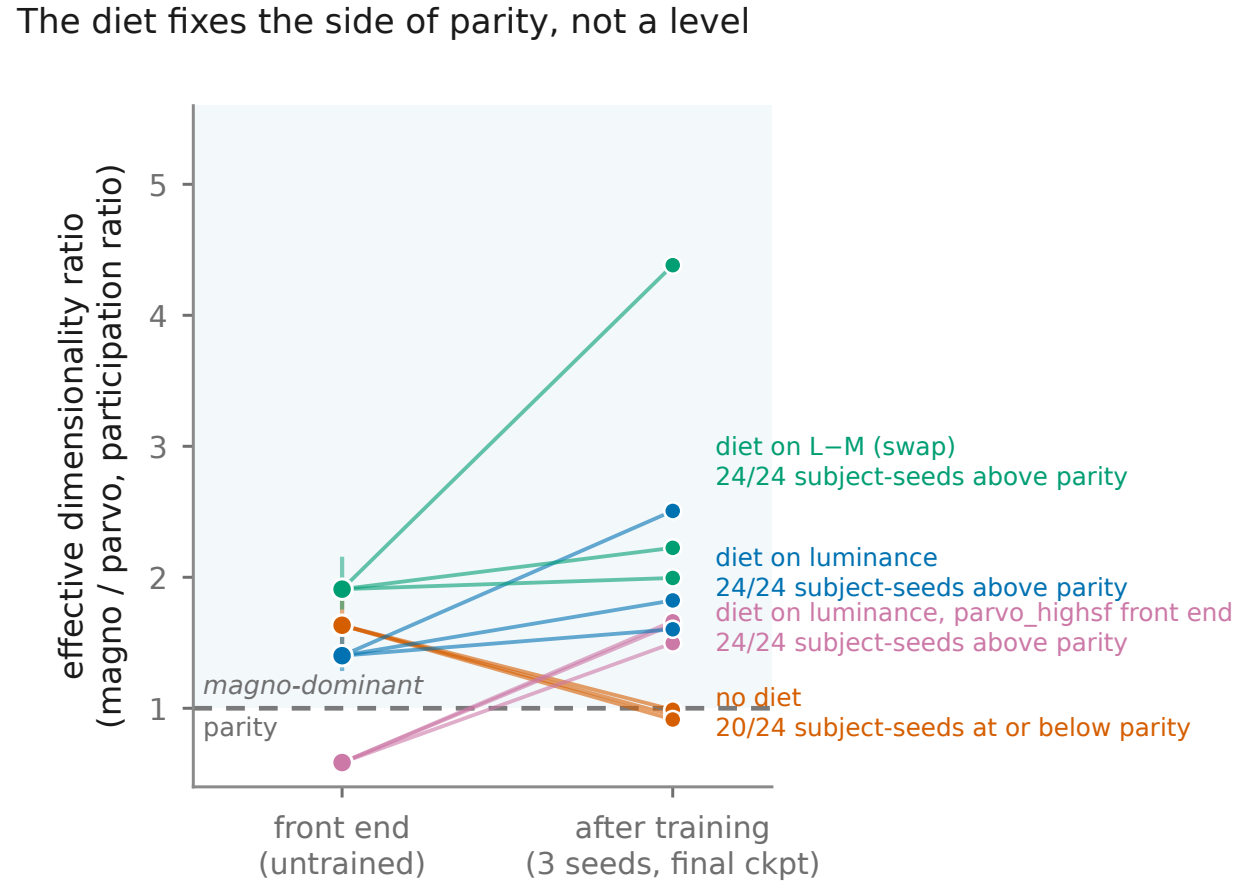
with a slope of 1.21 to 1.28), we read -0.834 as a sign-and-ordering result, not a calibrated effect size. The raw statistic per ROI is drawn beside the normalised one in Extended Data Fig. 3a; the diet arm's normalised index approaches -0.84 at hV4, near the index's bound, and the raw profile carries the same ordering without that bound.

Extended Data



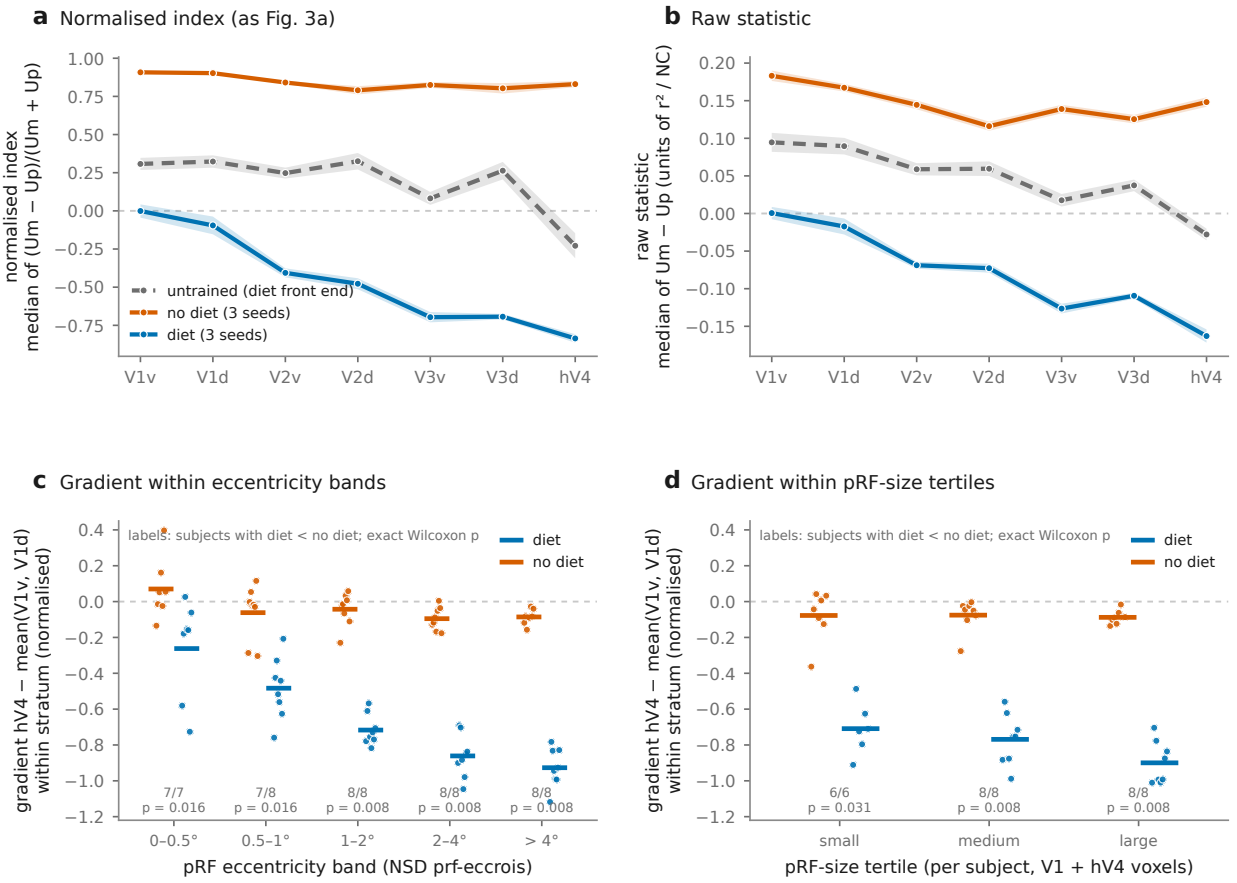
Extended Data Figure 1 | Out-of-distribution degradation curves behind Figure 5a.

Accuracy relative to each seed's own clean accuracy as a function of corruption severity, for seven corruption families (a–g) and the four front-end conditions, three seeds each (thin lines) with the seed mean (thick). Corruptions are applied to the RGB image before the front end, so each model meets the corruption through its own optics, and are scored in the models' native 112-class label space at each seed's best saved checkpoint. Five families were declared untrained-on before the run (uniform noise, high-pass, false colour, phase scrambling, rotation) and two as trained-in positive controls (contrast, low-pass), since the developmental transform is itself a low-pass and contrast-sensitivity manipulation. The diet-trained model degrades faster on uniform noise, high-pass, false colour and rotation, is indistinguishable from the control on phase scrambling, and is more robust only on low-pass. The two conditions that carry no shape bias (swap, `parvo_highsf`) track the no-diet control.



Extended Data Figure 2 | Effective dimensionality: the diet fixes which side of parity training ends on, not a level.

Ratio of the participation ratio of the magno stream's features to the parvo stream's, computed on 1,000 NSD images per subject (n = 8; Methods). Left: each arm's front end before any training, marker at the subject mean with the subject range as a whisker. Right: every training seed at the final checkpoint, each the mean over 8 subjects, fanned from the front end it started at. The four front ends hand the network very different asymmetries before learning (0.59 to 1.91, a 3.3-fold range), including one that is inverted: in `parvo_highsf` the parvo stream is the more dimensional stream in 8/8 subjects. After training every diet-trained seed ends magno-dominant (24/24 subject-seeds above parity in each of the three diet arms) while the no-diet arm collapses to parity (20/24 at or below). No level is labelled: the within-arm seed range (up to 2.39 at the final checkpoint, 0.79 at the best saved one) exceeds the gaps between arm means, so any statement of the form "training converges to X" is unsupported at three seeds. The `parvo_highsf` case also separates dimensionality from shape bias: its second stream is the more dimensional one while remaining texture-biased.

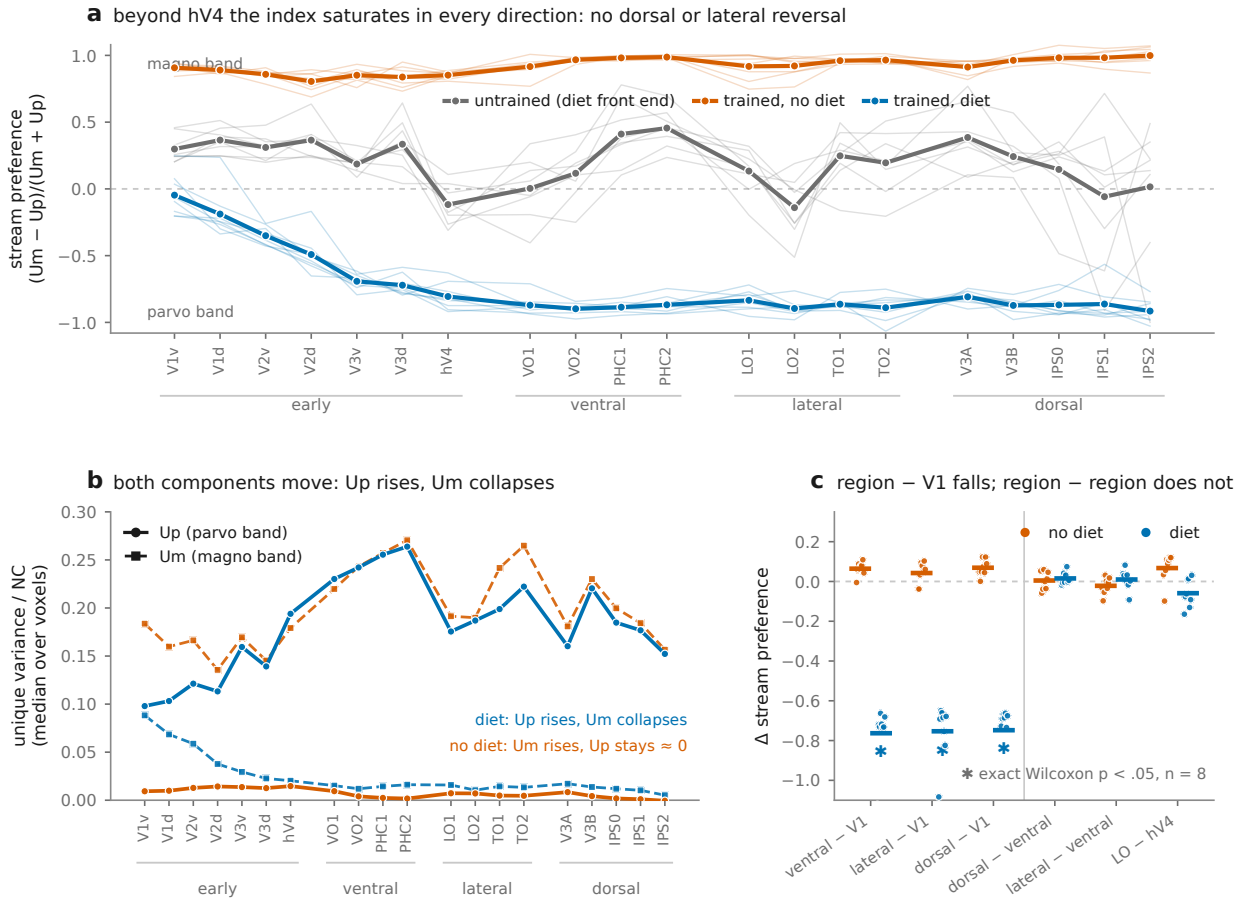


Extended Data Figure 3 | The cortical gradient against its nuisance axes: the raw statistic, eccentricity and receptive-field size.

(**a, b**) The stream-preference profile of Fig. 3a on both statistics: (a) the normalised per-voxel index (U_m − U_p)/(U_m + U_p), median over reliable voxels per ROI, and (b) the raw per-voxel difference U_m − U_p in units of noise-ceiling-normalised r². Lines are means over 8 subjects, shading ± SEM; the trained arms average three seeds per subject, the untrained network on the diet front end is a single model. The diet arm's normalised index approaches −0.84 at hV4, near the index's bound of −1; the raw statistic carries the same ordering (V1 near zero, hV4 at −0.16) without a bound, and the two-statistics rule of Methods 4.7 requires every claim to hold on both.

(**c**) The gradient hV4 − mean(V1v, V1d) computed within each of NSD's five eccentricity bands (`prf-eccrois`), so that V1 and hV4 voxels are compared only with voxels of matched eccentricity. One point per subject per band (seed-mean for the trained arms), bars = mean; a cell needs at least 20 voxels per ROI or the subject is dropped from that band. In the foveal band this drops subject 02 in every arm, whose hV4 has 19 reliable foveal voxels, and leaves subject 04's V1 value resting on V1v alone, its V1d having 5 (the voxel selection is identical across arms, so the same subject is missing from the diet and the no-diet points). Under the diet the gradient is negative in every band (6/7 subjects in the foveal band, 8/8 in the other four) and grows with eccentricity, from −0.26 in the foveal band to −0.93 beyond 4°; without the diet it stays between −0.10 and +0.07 in every band. Labels give the number of subjects in which the diet arm is more negative than the no-diet arm and the exact Wilcoxon p for that band. Eccentricity therefore modulates the diet arm's gradient but does not produce it.

(**d**) The same gradient within pRF-size tertiles, boundaries set per subject on the pooled V1 + hV4 voxels using NSD's pRF size estimates (voxels with pRF $R^2 > 5\%$). Under the diet the gradient is negative in every tertile (−0.71, −0.77, −0.90; all subjects), without the diet it is near zero in every tertile (−0.08 each), so the hierarchical contrast is not a receptive-field-size contrast in disguise. The within-subject regressions of Extended Data Table 9 give the same picture on the continuous predictors.

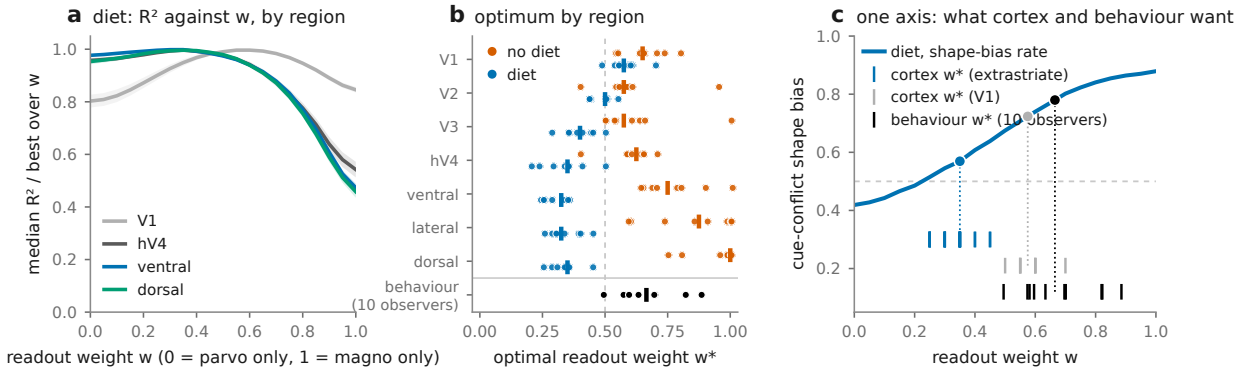


Extended Data Figure 4 | Beyond hV4 the gradient saturates in every direction: the texture-biased stream carries extrastriate cortex whole, ventral, lateral and dorsal alike.

(**a**) Stream-preference index, the median over reliable voxels of (Um − Up)/(Um + Up) (noise ceiling > 0.1, Um + Up > 10^{-3}, ≥ 20 voxels per ROI per subject), on the 22 ROIs of the Wang et al. (2015) atlas that pass the voxel rule in NSD's nsdgeneral mask, grouped as early (V1–hV4), ventral (VO1/2, PHC1/2), lateral (LO1/2, TO1/2) and dorsal (V3A/B, IPS0–2). Thick lines are the mean over the eight subjects, thin lines individual subjects; the trained arms are the mean over three training seeds per subject, the untrained network is one seed on the diet front end. Under the diet the index falls from −0.05 in V1v to −0.81 in hV4 and then sits between −0.80 and −0.94 in every extrastriate ROI. Without the diet the index is +0.80 to +1.00 everywhere. The same saturation holds on NSD's stream atlas and on the face-, place-, body- and word-selective fLoc ROIs (Extended Data Table 11).

(**b**) The two components of the index, both normalised by the noise ceiling: unique variance of the parvo band (Up, circles, solid) and of the magno band (Um, squares, dashed), median over the same voxels. Under the diet Up rises monotonically from 0.10 (V1v) to 0.26 (PHC2) while Um collapses from 0.09 to about 0.01 beyond V3; without the diet the luminance stream shows the same rise (0.18 to 0.27) and the L−M stream's unique variance stays near 0.01 throughout. The rise toward higher areas therefore belongs to whichever stream carries the full-resolution image; the diet moves it from the luminance stream to the chromatic one.

(**c**) Regional contrasts per subject (points) with the mean as a bar. Left of the divider: each region's mean index minus the V1 mean, which is large and negative under the diet in all three directions (ventral −0.76, lateral −0.75, dorsal −0.75; 8/8 subjects, exact Wilcoxon p = .0078) and small and positive without it. Right of the divider: region-to-region contrasts (dorsal − ventral, lateral − ventral, LO − hV4), none of which departs from zero under either arm (all $|\Delta| < 0.06$, $p > .1$). The model reproduces no magnocellular preference of dorsal cortex and no shape-selective reversal in LO.



Extended Data Figure 5 | The readout weight cortex prefers is not the one behaviour prefers.

The classifier's input is $z(w) = w \cdot y_1 + (1 - w) \cdot y_2$, where y_1 and y_2 are the 512-dimensional outputs of the magno and parvo streams (Methods 4.4). Results 2.4 sweeps w and reads the cue-conflict rate; here the same $z(w)$, for w from 0 to 1 in steps of 0.05, is the design of a ridge encoding model of every voxel in NSD's general visual mask (same split, grouped folds and alpha grid as the banded model; Methods 4.5). Because $z(w)$ sums the streams dimension by dimension, the ridge cannot undo the mixture: w constrains the design rather than re-parameterising it.

(**a**) Diet arm, three seeds averaged per subject: median test r^2 over an ROI's reliable voxels (noise ceiling > 0.1) as a function of w, normalised by its maximum over w, mean ± s.e.m. over eight subjects. V1 peaks near the balanced readout; hV4, ventral cortex (VO1/2, PHC1/2) and dorsal cortex (V3A/B, IPS0–2) peak on the parvo side. The curves are shallow: at the paper's w = 0.5 every ROI is within 3% of its best r^2, so the result is an ordering of optima, not a large loss.

(**b**) The optimum w* per subject (points) and its median (bar) for each region, both arms. Under the diet (blue) w* falls monotonically down the hierarchy, from 0.58 in V1 to 0.50, 0.40 and 0.35 in V2, V3 and hV4 and to 0.33–0.35 in every extrastriate group; the extrastriate mean is below V1 in 8/8 subjects ($\Delta = -0.25$, exact Wilcoxon $p = .0078$) and the Spearman correlation between hierarchical rank and w* over V1–hV4 is ≤ -0.80 in every subject. Without the diet (vermillion) the optimum runs the other way, from 0.65 in V1 to 0.75–1.00 beyond hV4 (7/8 subjects, $p = .023$), as the luminance stream carries the image. The bottom row is the behaviour-optimal weight: the w that best reproduces each of ten human observers' trial-by-trial cue-conflict choices (trial fit of Fig. 4; diet final checkpoints, three seeds averaged), median 0.67, range 0.49–0.89. The extrastriate optima (0.27–0.40) and the observers' optima do not overlap (Mann–Whitney $p = .0004$).

(**c**) Both optima on the axis of the behavioural sweep: the diet network's cue-conflict shape bias as a function of w (mean of three seeds, final checkpoints). Tick marks are individual subjects (cortex) or observers (behaviour); circles mark the medians. Reading the network out with the weight extrastriate cortex prefers gives a rate of 0.56, barely above parity; the weight that best predicts human choices gives 0.78; the human rate of 0.96 lies beyond the sweep's reach. The two demands are met by the same frozen streams and differ only in w.